Ricardo Leite Martins Bazarin - Academic Documentation

Ricardo Leite Martins Bazarin

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Mechanical Engineer, graduated from the Federal Technological University of Paraná (UTFPR), Guarapuava/□□ campus. He holds both a master's degree and a doctorate in the Thermal Sciences area from the Graduate Program in Mechanical and Materials Engineering at UTFPR, Curitiba campus. He has participated in research groups on Porous Media and Rheology, where he developed work in numerical modeling applied to Thermal Sciences and Fluid Mechanics, with emphasis on the use and development of the Lattice Boltzmann Method. Currently, he is a researcher at the Federal University of Santa Catarina (UFSC), specializing in digital rock simulation using the Lattice Boltzmann Method on High-Performance Computing (HPC) systems.

- □ Curriculum Lattes (Brazilian Academic Curriculum)
- □ GitHub Page
- □ Google Scholar

Articles Published in Periodics

- 2021-Boundary Effects on the Tortuosity and Permeability of Idealized Porous Media
  - Authors: Bazarin, De Lai, Naaktgeboren, Junqueira.
  - Area: (Porous media fluid-flow) Single-phase porous properties characterization in fractal porous media
- 2021-Moments-based method for boundary conditions in the lattice Boltzmann framework: A comparative analysis for the lid driven cavity flow
  - Authors: Bazarin, Philippi, Randles, Hegele Jr.
  - Area: (Boundary conditions modeling)
- 2024-Improved lattice Boltzmann model for immiscible multicomponent systems with high viscosity gradients at the interface
  - Authors: Bazarin, Naaktgeboren, Junqueira, Philippi, Hegele Jr.
  - Area: (Multicomponent/multiphase modeling) Numerical modeling for multicomponent fluid-flow
- 2025-Lattice Boltzmann method assessment for the Buckley–Leverett equation: Extension to dynamic capillary effect
  - Authors: Bazarin, dos Santos, Siebert.
  - Area: (Porous media fluid-flow - Multicomponent/multiphase modeling) Numerical modeling of multicomponent/multiphase flow in porous-continuous porous media.

- Suplemental Materials - C++ code of lattice Boltzmann simulation presented.

Academic Material

The full table of contents is generated automatically from the `_toc.yml` file and appears in the sidebar (HTML) or in the table of contents (PDF).

Part I

Lattice Boltzmann Method - Application to Different Partial Differential Equation

CHAPTER

ONE

DIFFUSIVE EQUATION

The diffusive equation is described by

$$\partial_t \phi - \partial_\alpha \left(\nu \partial_\alpha \phi \right) = 0 \tag{1.1}$$

where $\phi(\alpha, t)$ is a time and space dependent parameter and ν is diffusive coeficient.

1.1 Numerical Discretization

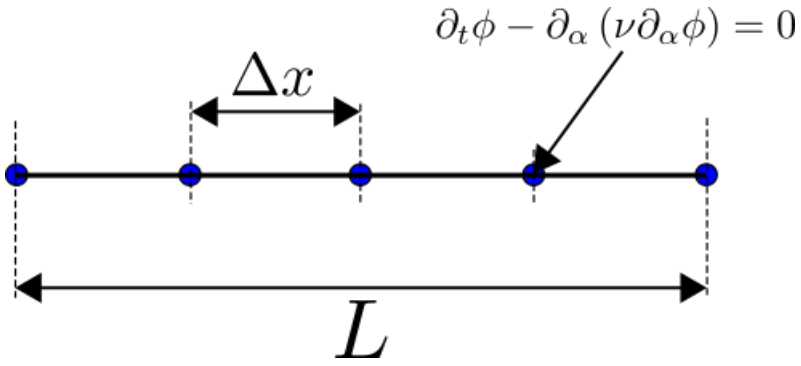


Fig. 1.1: 1D regular grid example for diffusive equation.

1.1.1 FDM Discretization

Discretizing the Eq. (1.1) through finite diference method, we have for forward time and central space (FTCS) discretizations:

$$\partial_t \phi - \partial_\alpha \left(\nu \partial_\alpha \phi \right) = \frac{\phi(x, t + \Delta_t) - \phi(x, t)}{\Delta_t} - \nu \frac{\phi(x - \Delta_x, t) - 2\phi(x, t) + \phi(x + \Delta_x, t)}{\Delta_x^2} = 0,$$

for ν constant and isolating the term $u(x, t + \Delta_t)$, we can describe the ϕ time evolution:

$$\phi(x, t + \Delta_t) = \phi(x, t) + \frac{\nu \Delta_t}{\Delta_x^2} \left(\phi(x - \Delta_x, t) - 2\phi(x, t) + \phi(x + \Delta_x, t) \right).$$

1.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation (BGK-LB):

$$f_i(x_\alpha + c_{i,\alpha}\delta t, t + \delta t) = f_i(x_\alpha, t) - \left(\frac{f_i - f_i^{eq}}{\tau}\right), \tag{1.2}$$

where the equilibrium distribution function is defined by

$$f_i^{eq} = w_i \phi(x_\alpha, t),$$

and the equilibrium moments are given by

$$\sum_{i=0} f_i^{eq} = \phi, \qquad \sum_i f_i^{eq} c_{i,\alpha} = 0 \qquad \text{and} \qquad \sum_i f_i^{eq} c_{i,\alpha} c_{i,\beta} = c_s^2 \phi \delta_{\alpha\beta}. \tag{1.3}$$

Through Chapman–Enskog analysis, it is demonstrated that the first-order non-equilibrium moment describes the pressure gradient and, consequently, the average velocity:

$$m_\alpha^{neq} = \sum_i c_{i,\alpha}\left(f_i - f_i^{eq}\right) = -(c_s^2 \tau \delta_t)\partial_\alpha \phi \qquad \text{and} \qquad \tau = \frac{\nu}{c_s^2 \delta_t} + \frac{1}{2}.$$

Lattice Direction Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i = \phi}_{\text{Zero-Order Moment}}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} = 0, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Moment}} = 0,$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} = \frac{\phi}{3}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} = 0 \qquad and \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} = \frac{\phi}{3}$$

Chapmann-Enskog Analysis

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i\left(x + e_i \Delta t, t + \Delta t\right)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i\left(x_\alpha + c_{i,\alpha}\Delta t, t + \Delta t\right) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = -\left(\frac{f_i - f_i^{eq}}{\tau}\right). \tag{1.4}$$

Rescaling the dimensionless form of the Eq. (1.4) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = -\frac{1}{Kn}\left(\frac{f_i - f_{eq,i}}{\tau}\right) \qquad \rightarrow \qquad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = -\frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \cdots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn\partial_t^{(1)} + Kn^2\partial_t^{(2)} + \cdots$), and separating the equation in orders up to the order Kn^2:

$$
\begin{aligned}
&(Kn^{(0)}): \quad f_i^{(0)} = f_i^{eq}, \\
&(Kn^{(1)}): \quad \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = -\frac{f_i^{(1)}}{\tau}, \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(0)}}{2} = -\frac{f_i^{(2)}}{\tau}, \\
&\text{ou} \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)\left(1 - \frac{1}{2\tau}\right) f_i^{(1)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(1)} \text{ Formulation} \\
&\text{ou} \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2\left(\frac{1}{2} - \tau\right) f_i^{(0)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(0)} \text{ Formulation.}
\end{aligned}
\tag{1.5}
$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (1.5) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$
\begin{aligned}
&(Kn^{(1)}): \sum_i \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = \sum_i \left(-\frac{f_i^{(1)}}{\tau}\right), \\
&(Kn^{(1)}): \partial_t^{(0)}\phi = 0,
\end{aligned}
$$

and

$$
\begin{aligned}
&(Kn^{(2)}): \sum_i \left(\partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2\left(\frac{1}{2} - \tau\right) f_i^{(0)}\right) = \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
&(Kn^{(2)}): \partial_t^{(1)}\phi + \left(\frac{1}{2} - \tau\right)\partial_\alpha\partial_\beta\left(c_s^2\phi\delta_{\alpha\beta}\right) = 0.
\end{aligned}
$$

or

$$
\begin{aligned}
&(Kn^{(2)}): \quad \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)}\right] = \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
&(Kn^{(2)}): \quad \partial_t^{(1)}\phi + \left(1 - \frac{1}{2\tau}\right) m_\alpha^{(1)} = 0.
\end{aligned}
$$

where $m_\alpha^{(1)}$ is the first-order moment of $f_i^{(1)}$ and $\sum_{i=1} f_i^{(j)} = 0$ for $j \geq 1$ due to imposition of ϕ conservation. To compute the moment $m_\alpha^{(1)}$, we multiply Eq. (1.5) by $c_{i,\alpha}$ and sum over all lattice directions:

$$
\begin{aligned}
&(Kn^{(1)}): \sum_i c_{i,\alpha}\left(\partial_t^{(0)} + e_{\beta,i}\partial_\beta\right) f_i^{(0)} = \sum_i c_{i,\alpha}\left(-\frac{f_i^{(1)}}{\tau}\right), \\
&(Kn^{(1)}): m_\alpha^{(1)} = -\tau\partial_\beta\left(c_s^2\phi\delta_{\alpha\beta}\right).
\end{aligned}
\tag{1.6}
$$

By substituting Eq. (1.6) into Eq. (1.6), we recover the Eq. (1.1), where $\nu = c_s^2(\tau - 1/2)$. In the regularized formulation of the lattice Boltzmann equation, the first-order correction is approximated as $f_i^{(1)} \approx (f_i - f_i^{(0)}) = f_i^{neq}$, where higher-order Knudsen moments are filtered out and the collision term is reformulated in terms of f_i^{neq}.

Boundary Conditions

The boundary conditions for the lattices can be derived by solving a linear system of known moments.

D1Q2:	Boundaries	Layers	
		West	East
	Unknown f_i	$f_2 = \phi_e - f_1$	$f_1 = \phi_w - f_2$
D1Q3:	Boundaries	Layers	
		West	East
	Unknown f_i	$f_2 = \phi_e - f_1 - f_0$	$f_1 = \phi_w - f_2 - f_0$

1.2 1st Benchmark: Single sine mode on a finite rod (Dirichlet)

PDE: $(\phi_t = \partial_x(\nu\partial_x\phi), (0 < x < L,\ t > 0)$

BC: $(\phi(0,t) = \phi(L,t) = 0)$

IC: $\phi(x,0) = \sin\left(\frac{\pi x}{L}\right)$

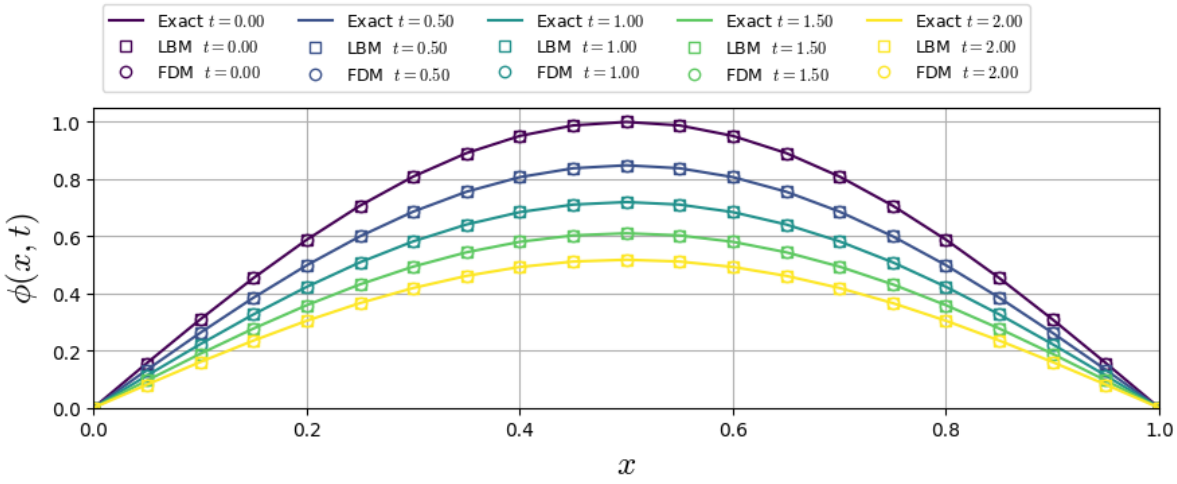
Solution

$$\phi(x,t) = \sin\left(\frac{\pi x}{L}\right)\exp\left(-\nu\frac{\pi^2}{L^2}t\right).$$

1.2.1 LBM D1Q3 Solution ($\tau = 1$)

1.2.2 FDM Solution

1.2.3 Numerical and Analytical Comparisson



CHAPTER

TWO

POISSON EQUATION (STEADY-STATE DIFFUSIVE EQUATION)

The Poisson is partial differential equation employed to represent problems as the pressure distribution over steady-state in space. It is characterized by a steady-state version of the difusive equation, i.e, the parameter is independent of time. The partial differential equation is given by

$$\partial_\alpha \left(\nu \partial_\alpha p\right) = 0 \tag{2.1}$$

where p is the pressure and ν diffusive control paremeter.

2.1 Numerical Discretization

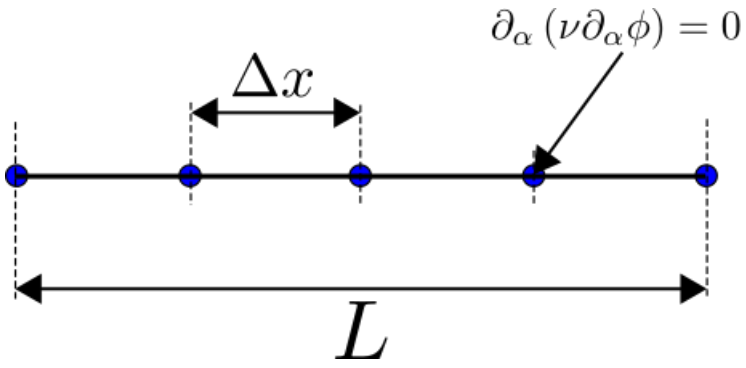


Fig. 2.1: 1D regular grid example for diffusive equation.

2.1.1 FDM Discretization

Discretizing the Eq. (1.1) through finite diference method, we have for forward time and central space (FTCS) discretizations:

$$\partial_\alpha \left(\nu \partial_\alpha p\right) = \nu \frac{p(x - \Delta_x, t) - 2p(x, t) + p(x + \Delta_x, t)}{\Delta_x^2} = 0,$$

for ν constant and isolating the term $u(x, t + \Delta_t)$, we can describe the p time evolution:

$$p(x, t) = \frac{p(x - \Delta_x, t) + p(x + \Delta_x, t)}{2}.$$

2.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation (BGK-LB) for the lattice D1Q3:

$$f_i(x_\alpha + c_{i,\alpha}\delta t, t' + \delta t) = f_i(x_\alpha, t') - \left(\frac{f_i - f_i^{eq}}{\tau}\right), \tag{2.2}$$

the equilibrium distribution function is defined by

$$f_i^{eq} = \begin{cases} (w_0 - 1)\,\hat{p}(x_\alpha, t') & i = 0 \\ w_i \hat{p}(x_\alpha, t') & i \neq 0 \end{cases},$$

where $\hat{p}$ is a shifted pressure, shifted by a reference constant pressure p_0, i.e., $p = p_0 + \hat{p}$ (This assumption is necessary due to pressure by definition can not be negative, but negative pressure values can be found depending of initial simulation set). The equilibrium moments are given by

$$\sum_{i=0} f_i^{eq} = 0, \qquad \frac{1}{1-w_0}\sum_{i=1} f_i^{eq} = \frac{1}{1-w_0}\sum_{i=1} f_i = \hat{p}, \qquad \sum_i f_i^{eq} c_{i,\alpha} = 0 \qquad \text{and} \qquad \sum_i f_i^{eq} c_{i,\alpha} c_{i,\beta} = c_s^2 \hat{p}\delta_{\alpha\beta}. \tag{2.3}$$

Through Chapman–Enskog analysis, it is demonstrated that the first-order non-equilibrium moment describes the pressure gradient and, consequently, the average velocity:

$$m_\alpha^{neq} = \sum_i c_{i,\alpha}\,(f_i - f_i^{eq}) = -(c_s^2 \tau \delta_t)\partial_\alpha \hat{p} \qquad \text{and} \qquad \tau = \frac{\nu}{c_s^2 \delta_t} + \frac{1}{2}.$$

Lattice Direction Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i = 0}_{\text{Zero-Order Moment}}, \qquad \underbrace{\frac{1}{1-w_0}\sum_{i=1} f_i^{eq} = \frac{1}{1-w_0}\sum_{i=1} f_i = \hat{p}}_{\text{Zero-Order Moment}}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} = 0}_{\text{x-First-Order Moment}}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} = 0}_{\text{y-First-Order Moment}}$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} = \frac{\hat{p}}{3}}_{\text{xx-Second-Order Moment}}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} = 0}_{\text{xy-Second-Order Moment}} \qquad and \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} = \frac{\hat{p}}{3}}_{\text{yy-Second-Order Moment}}$$

Chapmann-Enskog Analysis

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i\,(x + e_i\Delta t, t + \Delta t)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i\left(x_\alpha + c_{i,\alpha}\Delta t, t + \Delta t\right) - f_i(x,t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = -\left(\frac{f_i - f_i^{eq}}{\tau}\right). \tag{2.4}$$

Rescaling the dimensionless form of the Eq. (2.4) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = -\frac{1}{Kn}\left(\frac{f_i - f_{eq,i}}{\tau}\right) \qquad \rightarrow \qquad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = -\frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \cdots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn\partial_t^{(1)} + Kn^2\partial_t^{(2)} + \cdots$), and separating the equation in orders up to the order Kn^2:

$$
\begin{aligned}
&(Kn^{(0)}): \quad f_i^{(0)} = f_i^{eq}, \\
&(Kn^{(1)}): \quad \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = -\frac{f_i^{(1)}}{\tau}, \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(0)}}{2} = -\frac{f_i^{(2)}}{\tau}, \\
&\text{ou} \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)\left(1 - \frac{1}{2\tau}\right) f_i^{(1)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(1)} \text{ Formulation} \\
&\text{ou} \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2\left(\frac{1}{2} - \tau\right) f_i^{(0)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(0)} \text{ Formulation.}
\end{aligned}
\tag{2.5}
$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (2.5) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$
\begin{aligned}
&(Kn^{(1)}): \sum_i \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = \sum_i \left(-\frac{f_i^{(1)}}{\tau}\right), \\
&(Kn^{(1)}): 0 = 0,
\end{aligned}
$$

and

$$
\begin{aligned}
&(Kn^{(2)}): \sum_i \left(\partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2\left(\frac{1}{2} - \tau\right) f_i^{(0)}\right) = \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
&(Kn^{(2)}): \left(\frac{1}{2} - \tau\right)\partial_\alpha\partial_\beta\left(c_s^2\hat{p}\delta_{\alpha\beta}\right) = 0.
\end{aligned}
$$

or

$$
\begin{aligned}
&(Kn^{(2)}): \quad \sum_i \left[\partial_t^{(1)} f_i^{(0)} - \left(1 - \frac{1}{2\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)}\right] = \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
&(Kn^{(2)}): \quad -\left(1 - \frac{1}{2\tau}\right) m_\alpha^{(1)} = 0.
\end{aligned}
$$

where $m_\alpha^{(1)}$ is the first-order moment of $f_i^{(1)}$ and $1/(1-w_0)\sum_{i=1} f_i^{(j)} = 0$ for $j \geq 1$ due to imposition of $\hat{p}$ conservation. To compute the moment $m_\alpha^{(1)}$, we multiply Eq. (2.5) by $c_{i,\alpha}$ and sum over all lattice directions:

$$
\begin{aligned}
&(Kn^{(1)}): \sum_i c_{i,\alpha}\left(\partial_t^{(0)} + e_{\beta,i}\partial_\beta\right) f_i^{(0)} = \sum_i c_{i,\alpha}\left(-\frac{f_i^{(1)}}{\tau}\right), \\
&(Kn^{(1)}): m_\alpha^{(1)} = -\tau\partial_\beta\left(c_s^2\hat{p}\delta_{\alpha\beta}\right).
\end{aligned}
\tag{2.6}
$$

By substituting Eq. (2.6) into Eq. (2.6), we recover the Eq. (2.1), where $\nu = c_s^2(\tau - 1/2)$. In the regularized formulation of the lattice Boltzmann equation, the first-order correction is approximated as $f_i^{(1)} \approx (f_i - f_i^{(0)}) = f_i^{neq}$, where higher-order Knudsen moments are filtered out and the collision term is reformulated in terms of f_i^{neq}.

Boundary Conditions

The boundary conditions for the lattice D2Q5 can be derived by solving a linear system of known moments, as illustrated in appendix/bcs-D2Q5-poisson-equation. For the D2Q5 lattice, the unknown distribution functions at each boundary are determined as summarized in Table below.

Boundaries	Layers	
	North	South
Unknown f_i	$f_2 = \hat{p}_n(1-w_0) - (f_1+f_3+f_4)$	$f_4 = \hat{p}_s(1-w_0) - (f_1+f_2+f_3)$
	East	West
	$f_3 = \hat{p}_e(1-w_0) - (f_1+f_2+f_4)$	$f_1 = \hat{p}_w(1-w_0) - (f_2+f_3+f_4)$
Boundaries	Concave Corners	
	Northwest	Northeast
Unknown f_i	$f_1 = -(\partial_x\hat{p} + \partial_y\hat{p})\tau c_s^2/2 - f_2 + \hat{p}_{nw}/3$	$f_3 = (\partial_x\hat{p} - \partial_y\hat{p})\tau c_s^2/2 - f_2 + \hat{p}_{ne}/3$
	$f_4 = (\partial_x\hat{p} + \partial_y\hat{p})\tau c_s^2/2 - f_3 + \hat{p}_{nw}/3$	$f_4 = -(\partial_x\hat{p} - \partial_y\hat{p})\tau c_s^2/2 - f_1 + \hat{p}_{ne}/3$
	Southwest	Southeast
	$f_1 = -(\partial_x\hat{p} - \partial_y\hat{p})\tau c_s^2/2 - f_4 + \hat{p}_{sw}/3$	$f_2 = -(\partial_x\hat{p} + \partial_y\hat{p})\tau c_s^2/2 - f_1 + \hat{p}_{se}/3$
	$f_2 = (\partial_x\hat{p} - \partial_y\hat{p})\tau c_s^2/2 - f_3 + \hat{p}_{sw}/3$	$f_3 = (\partial_x\hat{p} + \partial_y\hat{p})\tau c_s^2/2 - f_4 + \hat{p}_{se}/3$

2.2 1st Benchmark: Single sine mode on a finite rod (Dirichlet)

PDE: $(\phi_t = \partial_x(\nu\partial_x\phi), (0 < x < L,\ t > 0)$

BC: $(\phi(0,t) = \phi(L,t) = 0)$

IC: $\phi(x,0) = \sin\left(\frac{\pi x}{L}\right)$

Solution

$$\phi(x,t) = \sin\left(\frac{\pi x}{L}\right)\exp\left(-\nu\frac{\pi^2}{L^2}t\right).$$

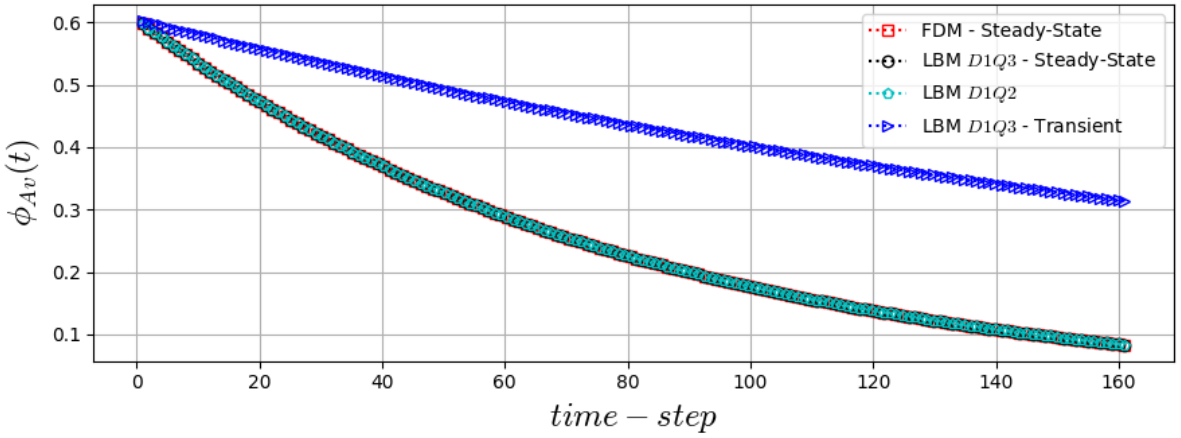
2.2.1 FDM Solution:

2.2.2 LBM D1Q3 Solution With Steady-Scheme:

2.2.3 LBM D1Q2 Solution (Not Accept the Steady-Scheme):

2.2.4 LBM D1Q3 Solution (Transient):

2.2.5 Numerical and Analytical Comparisson:



2.3 2nd Benchmark: 2D Poisson Pressure Field

Analyzing the accuracy of Poisson equation solver, we apply it to a two-dimensional analytical case. The geometry is described by a square domain of length L initialized with a constant dimensionless pressure $\hat{p}(x, y, t = 0) = 0$. The boundary conditions of the geometry are given by

$$\hat{p}(x = 0, y, t) = \frac{sinh\,(\pi(1 - y))}{sinh\,(\pi)}, \qquad \hat{p}(x = L, y, t) = -\frac{sinh\,(\pi(1 - y))}{sinh\,(\pi)},$$

$$\hat{p}(x, y = 0, t) = cos(\pi x), \qquad \hat{p}(x, y = L, t) = 0.$$

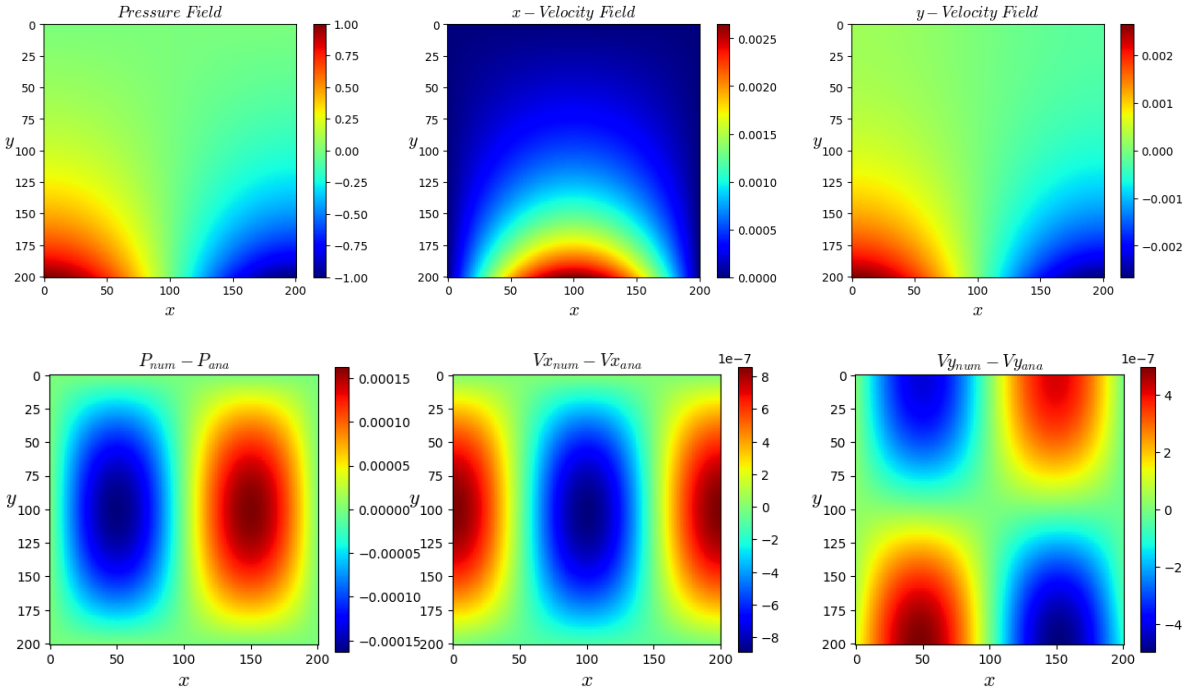
The pressure analytical solution for Eq. (2.1) pressure is given

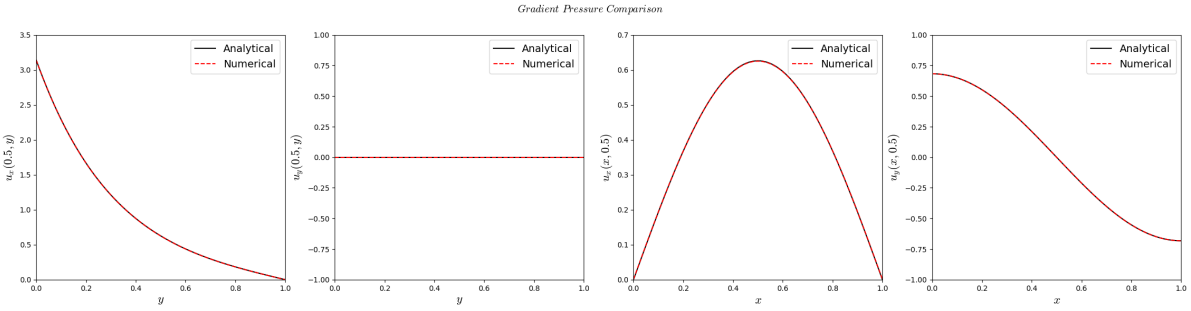
$$\hat{p}(x, y, t) = cos(\pi x)\frac{sinh\,(\pi(1 - y))}{sinh\,(\pi)}$$

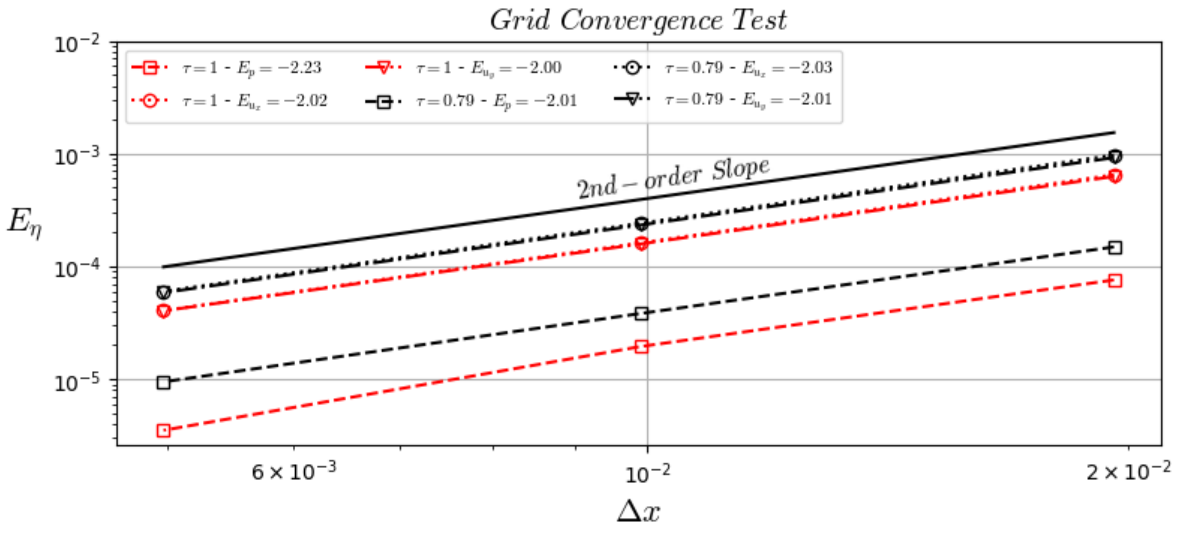
where Dp_0 is considered equal to 1, consequently, velocity values can be analytically obtained by the equation:

$$u_x = D\partial_x p = \pi sin(\pi x)\frac{sinh\,(\pi(1 - y))}{sinh\,(\pi)}, \qquad u_y = D\partial_y p = \pi cos(\pi x)\frac{cosh\,(\pi(1 - y))}{sinh\,(\pi)}.$$

2.3.1 LBM D2Q5 Code







CHAPTER

THREE

CONVECTIVE EQUATION (TRANSPORT EQUATION)

The convective equation also known as transport equation is a partial diferential equation (PDE) usually described by

$$\partial_t u + \partial_\alpha B_\alpha = 0 \tag{3.1}$$

where $u(\alpha, t)$ is a time and space dependent parameter that is being transporte and $B_\alpha(\alpha, t)$ is a second time and space dependent parameter that can also be depende of $u(\alpha, t)$ in non-linear convective diffusive equations.

3.1 Numerical Discretization

Discretizing the above equation in regular grid of 1D domain, we have

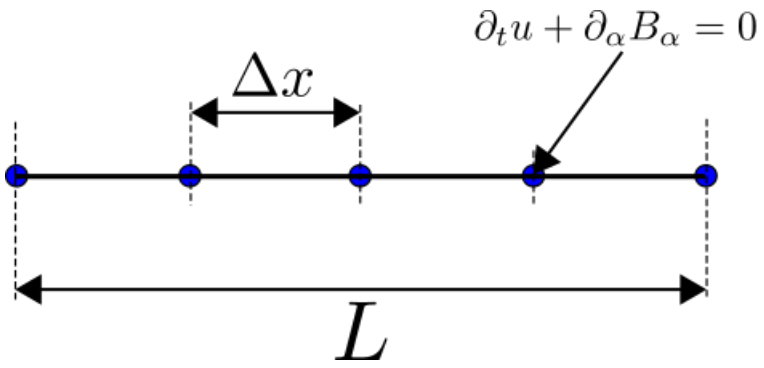


Fig. 3.1: 1D regular grid.

3.1.1 FDM Discretization

The Discretization through finite difference method can assume distinct forms that describe the problem with different accuracy, rate convergence and stability. In the sequence of this section, we will explore different variation that can work with the convective formulation.

Forward-Time and Forward-Space (FTFS)

Discretizing the Eq. (3.1) through finite diference method, we have for forward-time and forward-space (FTFS) discretizations:

$$\partial_t u + \partial_x B_\alpha = \frac{u(x,t+\Delta_t) - u(x,t)}{\Delta_t} + \frac{B_\alpha(x,t) - B_\alpha(x+\Delta_x,t)}{\Delta_x} = 0,$$

isolating the term $u(x,t+\Delta_t)$, we can describe the u time evolution:

$$u(x,t+\Delta_t) = u(x,t) - \frac{\Delta_t}{\Delta_x}\left(B_\alpha(x,t) - B_\alpha(x+\Delta_x,t)\right).$$

Forward-Time and Central-Space with Small Diffusivity (FTCS-SD)

Give that discretize convective equation with pure forwar-time and central-space (FTCS) can be unstable for several cases, employ a forwar-time and central-space with small diffusivity (FTCS-SD) is a alternative to improve stability and ensure a propertie conservation.

$$\begin{aligned}\partial_t u + \partial_x B_\alpha - \partial_\alpha\left(\nu\partial_\alpha u\right) = \quad & \frac{u(x,t+\Delta_t) - u(x,t)}{\Delta_t} + \frac{B_\alpha(x+\Delta_x,t) - B_\alpha(x-\Delta_x,t)}{2\Delta_x} \\ & -\nu\frac{u(x-\Delta_x,t) - 2u(x,t) + u(x+\Delta_x,t)}{\Delta_x^2} = 0,\end{aligned}$$

isolating the term $u(x,t+\Delta_t)$, we can describe the u time evolution:

$$u(x,t+\Delta_t) = u(x,t) + \frac{\Delta_t}{2\Delta_x}\left(B_\alpha(x-\Delta_x,t) - B_\alpha(x+\Delta_x,t)\right) + \frac{\nu\Delta_t}{\Delta_x^2}\left(u(x-\Delta_x,t) - 2u(x,t) + u(x+\Delta_x,t)\right).$$

3.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation, we assume a equilibrium state for the present problem, where $f_i = f_i^{eq}$ and the BGK lattice Boltzmann equation reduces to:

$$f_i\left(x_\alpha + c_{i,\alpha}\Delta t, t+\Delta t\right) - f_i(x_\alpha,t) = -\left(\frac{f_i - f_i^{eq}}{\tau}\right) \qquad \rightarrow \qquad f_i(x_\alpha + c_{i,\alpha}\delta t, t+\delta t) = f_i^{eq}, \tag{3.2}$$

where the equilibrium distribution function is defined by

$$f_i^{eq} = w_i\left(u + \frac{c_{i,\alpha}B_\alpha}{c_s^2}\right).$$

Lattice Direction Moments

Raw Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i = u}_{\text{Zero-Order Moment}}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} = B_\alpha, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}c_{i,x}}_{\text{xx-Second-Order Moment}} = \frac{u}{3}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}c_{i,x}c_{i,x}}_{\text{xxx-Third-Order Moment}} = B_\alpha$$

Hermite Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Hermite Moment}} = u, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} = B_\alpha$$

$$\underbrace{\sum_{i=0} f_i^{eq}\left(c_{i,x}c_{i,x} - c_s^2\right)}_{\text{xx-Second-Order Hermite Moment}} = 0, \qquad \underbrace{\sum_{i=0} f_i^{eq}\left(c_{i,x}c_{i,x} - 3c_s^2\right)c_{i,x}}_{\text{xxx-Third-Order Hermite Moment}} = 0$$

Chapmann-Enskog Analysis (Considering Equilibrium State - Consequently Disregarding Inherited Diffusive Effects)

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i\left(x + e_i\Delta t, t + \Delta t\right) = f_i^{'}$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i^{'} - f_i(x,t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = -\left(\frac{f_i - f_i^{eq}}{\tau}\right) \overset{\text{Equilibrium State}}{\rightarrow} f_i^{'} = \sum_{j=0}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = f_i^{eq}(x,t). \tag{3.3}$$

Rescaling the dimensionless form of the Eq. (3.3) in terms of the Knudsen number (Kn), we have

$$\sum_{j=0}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = f_i^{eq}(x_\alpha, t),$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Knf_i^{(1)} + Kn^2 f_i^{(2)} + \cdots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn\partial_t^{(1)} + Kn^2\partial_t^{(2)} + \cdots$), and separating the equation in orders up to the order Kn^2:

$$\begin{aligned}
&(Kn^{(0)}): \quad f_i^{(0)} = f_i^{eq},\\
&(Kn^{(1)}): \quad \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = 0,\\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(0)}}{2} = 0,\\
&\text{ou}\\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)} = 0,\\
&(Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)}\left(\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)}\right) + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)}}{2}\\
&\qquad + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^3 f_i^{(0)}}{6} = 0,\\
&\text{ou}\\
&(Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)} = 0.
\end{aligned} \tag{3.4}$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (3.4) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$\begin{aligned}
&(Kn^{(1)}): \sum_i \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = 0,\\
&(Kn^{(1)}): \partial_t^{(0)} u + \partial_\alpha B_\alpha = 0,
\end{aligned} \tag{3.5}$$

$$\begin{aligned}
&(Kn^{(2)}): \quad \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)}\right] = 0,\\
&(Kn^{(2)}): \quad \partial_t^{(1)} u + \partial_\alpha m_\alpha^{(1)} = 0,
\end{aligned} \tag{3.6}$$

and

$$\begin{aligned}(Kn^{(3)}): &\quad \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)}\right] = 0, \\ (Kn^{(3)}): &\quad \partial_t^{(2)} u + \partial_\alpha m_\alpha^{(2)} = 0.\end{aligned} \tag{3.7}$$

By summing the Eqs. (3.5), (3.6), and (3.7), and also extending to higher orders of Knudsen, we have

$$\partial_t u + \partial_\alpha B_\alpha + \underbrace{\partial_\alpha \left(m_\alpha^{(1)} + m_\alpha^{(2)} + ...\right)}_{\text{Numerical Errors}} = 0,$$

where these numerical error can be related to the numerical error from the approximation of a derivative term by an expansion in Taylor series.

Chapmann-Enskog Analysis (Full)

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i\left(x + e_i\Delta t, t + \Delta t\right)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i\left(x_\alpha + c_{i,\alpha}\Delta t, t + \Delta t\right) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = -\left(\frac{f_i - f_i^{eq}}{\tau}\right). \tag{3.8}$$

Rescaling the dimensionless form of the Eq. (3.8) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = -\frac{1}{Kn}\left(\frac{f_i - f_{eq,i}}{\tau}\right) \qquad \rightarrow \qquad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = -\frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Knf_i^{(1)} + Kn^2 f_i^{(2)} + \cdots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn\partial_t^{(1)} + Kn^2\partial_t^{(2)} + \cdots$), and separating the equation in orders up to the order Kn^2:

$$\begin{aligned}
&(Kn^{(0)}): \quad f_i^{(0)} = f_i^{eq}, \\
&(Kn^{(1)}): \quad \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = -\frac{f_i^{(1)}}{\tau}, \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(0)}}{2} = -\frac{f_i^{(2)}}{\tau}, \\
&\text{ou} \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)\left(1 - \frac{1}{2\tau}\right) f_i^{(1)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(1)} \text{ Formulation} \\
&\text{ou} \\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2\left(\frac{1}{2} - \tau\right) f_i^{(0)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(0)} \text{ Formulation} \\
&(Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)}\left(\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)}\right) + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)}}{2} \\
&\qquad + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^3 f_i^{(0)}}{6} = -\frac{f_i^{(3)}}{\tau}, \\
&\text{ou} \\
&(Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau}\right)\partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)} = -\frac{f_i^{(3)}}{\tau},
\end{aligned} \tag{3.9}$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (3.9) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$\begin{aligned}(Kn^{(1)}) &: \sum_i \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = -\sum_i \left(\frac{f_i^{(1)}}{\tau}\right), \\ (Kn^{(1)}) &: \partial_t^{(0)} u + \partial_\alpha B_\alpha = 0,\end{aligned} \tag{3.10}$$

$$\begin{aligned}(Kn^{(2)}) &: \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)}\right] = -\sum_i \left(\frac{f_i^{(2)}}{\tau}\right), \\ (Kn^{(2)}) &: \partial_t^{(1)} u + \left(1 - \frac{1}{2\tau}\right)\partial_\alpha m_\alpha^{(1)} = 0,\end{aligned} \tag{3.11}$$

To determine the unknow term $m_\alpha^{(1)}$, we sum the Eq. (3.9) for $Kn^{(1)}$ over the first-order moment:

$$\begin{aligned}&(Kn^{(1)}) : \sum_i \left(\partial_t^{(0)} + c_{i,\beta}\partial_\beta\right) f_i^{(0)} c_{i,\alpha} = -\sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha}\right), \\ &(Kn^{(1)}) : \partial_t^{(0)} B_\alpha + \partial_\beta\left(u c_s^2\right)\delta_{\alpha\beta} = -\frac{m_\alpha^{(1)}}{\tau} \qquad \rightarrow \qquad m_\alpha^{(1)} = -\tau\left(\partial_t^{(0)} B_\alpha + \partial_\beta\left(u c_s^2\right)\delta_{\alpha\beta}\right).\end{aligned} \tag{3.12}$$

Extending the analysis up to $Kn^{(3)}$

$$\begin{aligned}(Kn^{(3)}) &: \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau}\right)\partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)}\right. \\ &\quad \left. + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)}\right] = -\sum_i \left[\frac{f_i^{(3)}}{\tau}\right], \\ (Kn^{(3)}) &: \partial_t^{(2)} u + \partial_t^{(1)}\left(\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)}\right) + \partial_\alpha\partial_\beta m_{\alpha\beta}^{(1)} + \partial_\alpha m_\alpha^{(2)} = 0.\end{aligned} \tag{3.13}$$

By summing the Eqs. (3.10), (3.11), and (3.13), and also extending to higher orders of Knudsen, we have

$$\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha\left(\nu\partial_\alpha u\right) + \underbrace{\partial_\alpha m_\alpha^{(2)} + \partial_\alpha\partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(1)}\left(\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)}\right) - \left(\tau - \frac{1}{2}\right)\partial_t^{(0)}\partial_\alpha B_\alpha + ...}_{\text{Numerical Errors}} = 0,$$

where $\nu = c_s^2(\tau - 1/2)$. The numerical errors can be corrected using extra moments in the equilibrium distribution function and some ones can neglected due to its relevance decay exponentially with the decrease of Δx and Δt.

Boundary Conditions

The boundary conditions for the lattices can be derived by solving a linear system of known moments.

D1Q3:	Boundaries	Layers	
		West	East
	Unknown f_i	$f_2 = u_e - f_1 - f_0$	$f_1 = u_w - f_2 - f_0$

3.2 Benchmark - 1D Non-Linear Convective formulation of the Buckley-Leverett Equation

Considering a pure convective formulation of the Buckley-Leverett equation, where gravitational effects and capillary pressure are neglected, we have:

$$\partial_t(s_w) + \partial_\alpha \left[\frac{\langle u \rangle_\alpha}{\varphi} F(s_w) \right] = 0, \tag{3.14}$$

with the water fraction flow given by,

$$F(s_w) = \frac{s_w^2}{s_w^2 + M_\mu (1 - s_w)^2}. \tag{3.15}$$

The simulation parameters are chosen so that the average velocity component $\langle u \rangle_\alpha / \varphi$ equals $1\,\mathrm{m/s}$ and the dynamic viscosity ratio M_μ equals 1. We consider a one-dimensional porous medium of length $L_x = 20\,\mathrm{m}$, initially saturated with oil (see Fig. 3.2). Water is injected at the inlet $(x = 0)$ under a Dirichlet boundary condition $s_w(x = 0) = 1$. At the outlet $(x = L)$, a Neumann boundary condition specifying a zero gradient in water saturation is applied.

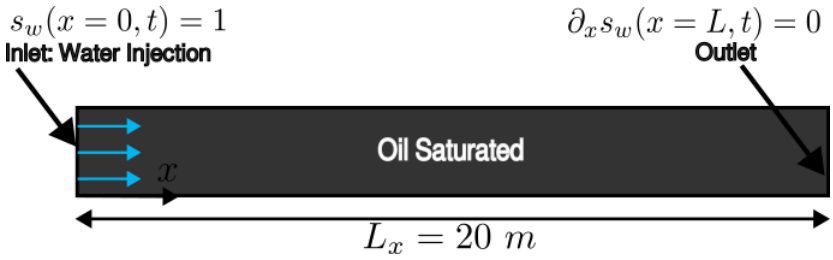


Fig. 3.2: Geometry of the non-linear convective problem: initial and boundary conditions.

3.2.1 Transient Analytical Solution

The non-linear convective problem given by the Eq. (3.14) has an analytical solution dependent of the initial condition (Riemann problem). In the present case, the solution for the specific initial condition is given by

$$s_w(x,t) = \begin{cases} tF' & \text{for} \quad x \leq \zeta t \\ \\ 0 & \text{for} \quad x > \zeta t \end{cases}$$

where $\zeta = F'(\tilde{s}_w) = F(\tilde{s}_w)/\tilde{s}_w$ is the shock wave speed, and $\tilde{s}_w$ is the sharp front saturation determined analytically by

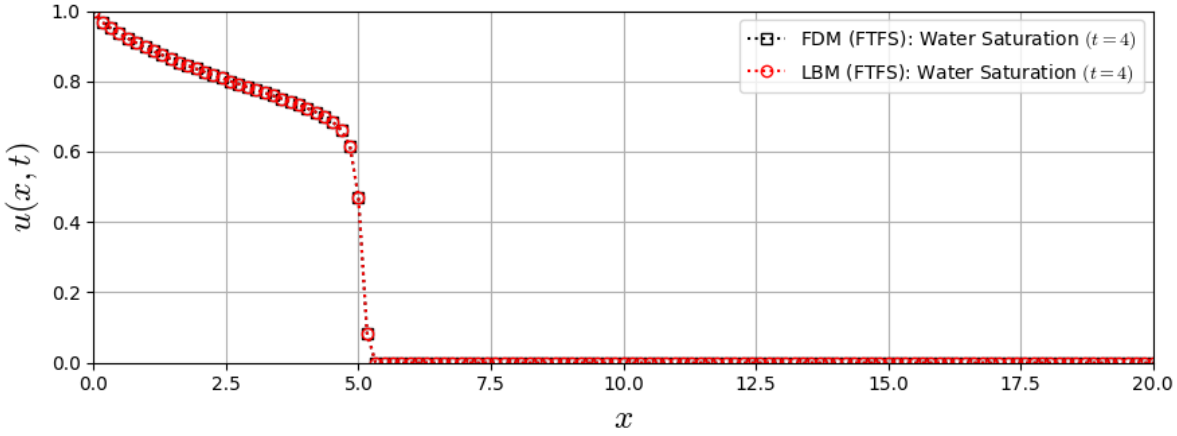
$$\tilde{s}_w = \frac{F(\tilde{s}_w)}{F'(\tilde{s}_w)} = \sqrt{\frac{M_\mu}{M_\mu + 1}}, \qquad \text{where} \qquad F^{'}(s_w) = \frac{dF}{ds_w} = \frac{2 s_w M_\mu (1 - s_w)}{(s_w^2 + M_\mu (1 - s_w)^2)^2}.$$

Forward Time and Forward Space - FTFS

FDM Solution:

Lattice Boltzmann Framework of FTFS:

Numerical and Analytical Comparisson

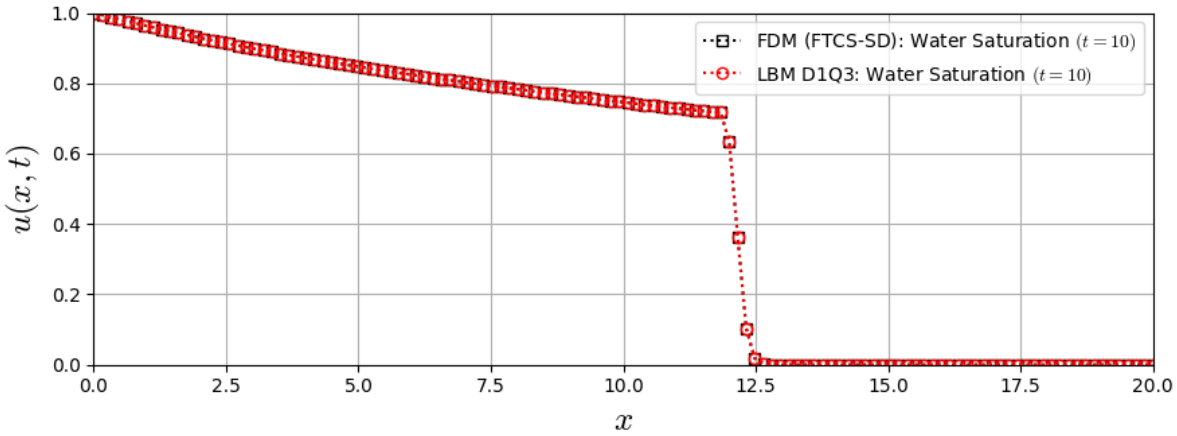


3.2.2 Forward-Time and Central-Space with Small Diffusivity (FTCS-SD)

Lattice Boltzmann Method (D1Q3):

FDM approach of LBM Framework D1Q3:

Numerical and Analytical Comparisson



CHAPTER

FOUR

CONVECTIVE-DIFFUSIVE EQUATION

The convective-diffusive equation is a partial diferential equation (PDE) commonly written as

$$\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha \left(\nu \partial_\alpha C\right) = 0 \tag{4.1}$$

where $u(\alpha, t)$ is a space and time dependent scalar parameter, $B_\alpha(\alpha, t)$ is a vector a C a scalar term, may be time and space dependent parameters and can also be also expressed as linear or non-linear functions of $u(\alpha, t)$. The term ν is the diffusion coefficient, a constituve term that controls the intensity of diffusive effect.

4.1 Numerical Discretization

Discretizing the above equation in regular grid of 1D domain, we have

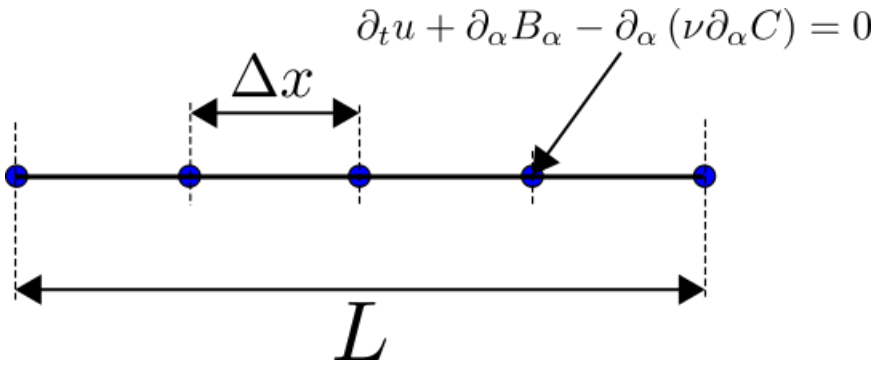


Fig. 4.1: 1D regular grid.

4.1.1 FDM Discretization

To explore different discretization approaches, we adopt forward-time, forward-space (commonly referred to as the upwind scheme), as well as central-difference schemes.

Forward-Time and Forward-Space (FTFSCS)

Discretizing the Eq. (4.1) through finite diference method, we have for forward-time, forward-space for advection and central-space for diffusion (FTFSCS) discretizations:

$$\begin{aligned}\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha \left(\nu \partial_\alpha C\right) = \quad & \frac{u(x,t+\Delta_t) - u(x,t)}{\Delta_t} + \frac{B_\alpha(x,t) - B_\alpha(x+\Delta_x,t)}{\Delta_x} \\ & -\nu \frac{C(x-\Delta_x,t) - 2C(x,t) + C(x+\Delta_x,t)}{\Delta_x^2} = 0,\end{aligned}$$

isolating the term $u(x,t+\Delta_t)$, we can describe the u time evolution:

$$u(x,t+\Delta_t) = u(x,t) - \frac{\Delta_t}{\Delta_x}\left(B_\alpha(x,t) - B_\alpha(x+\Delta_x,t)\right) + \frac{\nu\Delta_t}{\Delta_x^2}\left(C(x-\Delta_x,t) - 2C(x,t) + C(x+\Delta_x,t)\right)$$

Forward-Time and Central-Space with Small Diffusivity (FTCS)

Given the discretization of the convective-diffusive equation with pure forwar-time and central-space in advective and diffusive terms (FTCS), we have

$$\begin{aligned}\partial_t u + \partial_x B_\alpha - \partial_\alpha \left(\nu \partial_\alpha u\right) = \quad & \frac{u(x,t+\Delta_t) - u(x,t)}{\Delta_t} + \frac{B_\alpha(x+\Delta_x,t) - B_\alpha(x-\Delta_x,t)}{2\Delta_x} \\ & -\nu \frac{C(x-\Delta_x,t) - 2C(x,t) + C(x+\Delta_x,t)}{\Delta_x^2} = 0,\end{aligned}$$

isolating the term $u(x,t+\Delta_t)$, we can describe the u time evolution:

$$u(x,t+\Delta_t) = u(x,t) - \frac{\Delta_t}{2\Delta_x}\left(B_\alpha(x+\Delta_x,t) - B_\alpha(x-\Delta_x,t)\right) + \frac{\nu\Delta_t}{\Delta_x^2}\left(C(x-\Delta_x,t) - 2C(x,t) + C(x+\Delta_x,t)\right)$$

Forward-Time and Central-Space with Small Diffusivity (FTCS-SD)

Give that discretize convective equation with pure forwar-time and central-space (FTCS), employ a small diffusivity (FTCS-SD) is a alternative to improve stability and ensure a propertie conservation.

$$\begin{aligned}\partial_t u + \partial_x B_\alpha - \partial_\alpha \left(\nu_u \partial_\alpha u\right) - \partial_\alpha \left(\nu \partial_\alpha C\right) = \frac{u(x,t+\Delta_t) - u(x,t)}{\Delta_t} + \frac{B_\alpha(x+\Delta_x,t) - B_\alpha(x-\Delta_x,t)}{\Delta_x} - \\ \nu_u \frac{u(x-\Delta_x,t) - 2u(x,t) + u(x+\Delta_x,t)}{\Delta_x^2} - \nu \frac{C(x-\Delta_x,t) - 2C(x,t) + C(x+\Delta_x,t)}{\Delta_x^2} = 0,\end{aligned}$$

isolating the term $u(x,t+\Delta_t)$, we can describe the u time evolution:

$$\begin{aligned}u(x,t+\Delta_t) = \quad & u(x,t) - \frac{\Delta_t}{\Delta_x}\left(B_\alpha(x+\Delta_x,t) - B_\alpha(x-\Delta_x,t)\right) + \frac{\nu_u\Delta_t}{\Delta_x^2}\left(u(x-\Delta_x,t) - 2u(x,t) + u(x+\Delta_x,t)\right) \\ & + \frac{\nu\Delta_t}{\Delta_x^2}\left(C(x-\Delta_x,t) - 2C(x,t) + C(x+\Delta_x,t)\right).\end{aligned}$$

4.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation:

$$f_i\left(x_\alpha + c_{i,\alpha}\Delta t, t+\Delta t\right) - f_i(x_\alpha,t) = -\left(\frac{f_i - f_i^{eq}}{\tau}\right). \tag{4.2}$$

The equilibrium distribution function is defined by

$$f_i^{eq} = w_i \left(u + \frac{c_{i,\alpha} B_\alpha}{c_s^2} + \frac{\left(c_s^2 C \delta_{\alpha\beta} + \Lambda_{\alpha\beta} - c_s^2 u \delta_{\alpha\beta}\right)\left(c_{i,\alpha} c_{i,\beta} - c_s^2 \delta_{\alpha\beta}\right)}{2c_s^4} \right),$$

where $\Lambda_{\alpha\beta}$ is a correction term defined by

$$\Lambda_{\alpha\beta} = \int B_\alpha^{'} B_\beta^{'} du = \int \left(\frac{\partial B_\alpha}{\partial u} \frac{\partial B_\beta}{\partial u} \right) du.$$

Lattice Direction Moments

Raw Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Moment}} = u, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} = B_\alpha, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} = \frac{C}{3} + \Lambda_{\alpha\beta}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} = B_\alpha.$$

Hermite Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Hermite Moment}} = u, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} = B_\alpha,$$

$$\underbrace{\sum_{i=0} f_i^{eq} \left(c_{i,x} c_{i,x} - c_s^2\right)}_{\text{xx-Second-Order Hermite Moment}} = \frac{C}{3} + \Lambda_{\alpha\beta} - \frac{u}{3}, \qquad \underbrace{\sum_{i=0} f_i^{eq} \left(c_{i,x} c_{i,x} - 3c_s^2\right) c_{i,x}}_{\text{xxx-Third-Order Hermite Moment}} = 0.$$

Chapmann-Enskog Analysis

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i\left(x + e_i \Delta t, t + \Delta t\right)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i\left(x_\alpha + c_{i,\alpha}\Delta t, t + \Delta t\right) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = -\left(\frac{f_i - f_i^{eq}}{\tau} \right). \tag{4.3}$$

Rescaling the dimensionless form of the Eq. (4.3) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = -\frac{1}{Kn}\left(\frac{f_i - f_{eq,i}}{\tau} \right) \qquad \rightarrow \qquad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = -\frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \cdots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn\partial_t^{(1)} + Kn^2\partial_t^{(2)} + \cdots$), and separating the equation in orders up to the order Kn^2:

$$
\begin{aligned}
&(Kn^{(0)}): \quad f_i^{(0)} = f_i^{eq},\\
&(Kn^{(1)}): \quad \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = -\frac{f_i^{(1)}}{\tau},\\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(0)}}{2} = -\frac{f_i^{(2)}}{\tau},\\
&\text{ou}\\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)\left(1 - \frac{1}{2\tau}\right) f_i^{(1)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(1)} \text{ Formulation}\\
&\text{ou}\\
&(Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2\left(\frac{1}{2} - \tau\right) f_i^{(0)} = -\frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(0)} \text{ Formulation}\\
&(Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)}\left(\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)}\right) + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)}}{2}\\
&\qquad\qquad + \frac{\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^3 f_i^{(0)}}{6} = -\frac{f_i^{(3)}}{\tau},\\
&\text{ou}\\
&(Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau}\right)\partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)} + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)} = -\frac{f_i^{(3)}}{\tau},
\end{aligned}
\tag{4.4}
$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (4.4) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$
\begin{aligned}
&(Kn^{(1)}): \sum_i \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = -\sum_i \left(\frac{f_i^{(1)}}{\tau}\right),\\
&(Kn^{(1)}): \partial_t^{(0)} u + \partial_\alpha B_\alpha = 0,
\end{aligned}
\tag{4.5}
$$

$$
\begin{aligned}
&(Kn^{(2)}): \quad \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)}\right] = -\sum_i \left(\frac{f_i^{(2)}}{\tau}\right),\\
&(Kn^{(2)}): \quad \partial_t^{(1)} u + \left(1 - \frac{1}{2\tau}\right)\partial_\alpha m_\alpha^{(1)} = 0,
\end{aligned}
$$

To determine the unknow term $m_\alpha^{(1)}$, we sum the Eq. (4.4) for $Kn^{(1)}$ over the first-order moment:

$$
\begin{aligned}
&(Kn^{(1)}): \sum_i \left(\partial_t^{(0)} + c_{i,\beta}\partial_\beta\right) f_i^{(0)} c_{i,\alpha} = -\sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha}\right),\\
&(Kn^{(1)}): \partial_t^{(0)} B_\alpha + \partial_\beta\left(c_s^2 C\delta_{\alpha\beta} + \Lambda_{\alpha\beta}\right) = -\frac{m_\alpha^{(1)}}{\tau} \qquad \rightarrow \qquad m_\alpha^{(1)} = -\tau\partial_\beta\left(c_s^2 C\delta_{\alpha\beta}\right).
\end{aligned}
\tag{4.6}
$$

Extending the analysis up to $Kn^{(3)}$

$$
\begin{aligned}
&(Kn^{(3)}): \quad \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau}\right)\partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)}\right.\\
&\qquad\qquad \left. + \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(2)}\right] = -\sum_i \left[\frac{f_i^{(3)}}{\tau}\right],\\
&(Kn^{(3)}): \quad \partial_t^{(2)} u + \partial_t^{(1)}\left(\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)}\right) + \partial_\alpha\partial_\beta m_{\alpha\beta}^{(1)} + \partial_\alpha m_\alpha^{(2)} = 0.
\end{aligned}
\tag{4.7}
$$

By summing the Eqs. (4.5), (4.6), and (4.7), and also extending to higher orders of Knudsen, we have

$$\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha \left(\nu \partial_\alpha C\right) + \underbrace{\partial_\alpha m_\alpha^{(2)} + \partial_\alpha \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(1)} \left(\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)}\right) + ...}_{\text{Numerical Errors}} = 0,$$

where $\nu = c_s^2(\tau - 1/2)$. The numerical errors can be corrected using extra moments in the equilibrium distribution function and some ones can neglected due to its relevance decay exponentially with the decrease of Δx and Δt.

Boundary Conditions

The boundary conditions for the lattices can be derived by solving a linear system of known moments.

D1Q3:	Boundaries	Layers	
		West	East
	Unknown f_i	$f_2 = u_e - f_1 - f_0$	$f_1 = u_w - f_2 - f_0$

4.2 1st Benchmark: Decay of Sinoidal Wave

Considering a 1D **convection–diffusion equation** with constant coefficients $\$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2}, \qquad x \in \mathbb{R},\ t \geq 0,\$$ with **sinusoidal initial condition**

$$u(x, 0) = A\sin(kx + \phi),$$

where A is the initial amplitude, k the wavenumber, ϕ a phase shift.

Then the analytical solution is

$$\boxed{u(x,t) = Ae^{-\nu k^2 t}\sin\left(k(x - ct) + \phi\right)}$$

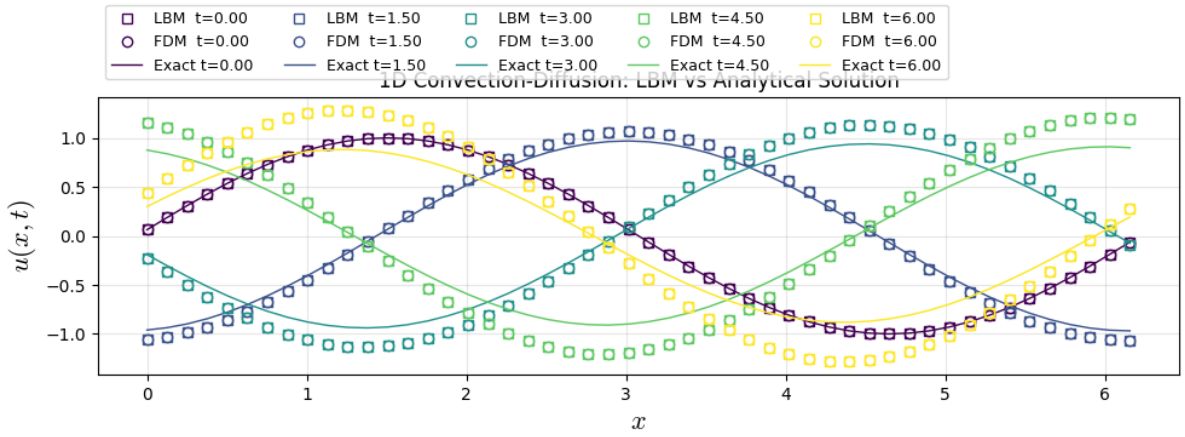
So:

- **Convection** shifts the wave: $x \mapsto x - ct$ (wave travels at speed c).
- **Diffusion** damps the amplitude exponentially: $A(t) = Ae^{-\nu k^2 t}$.

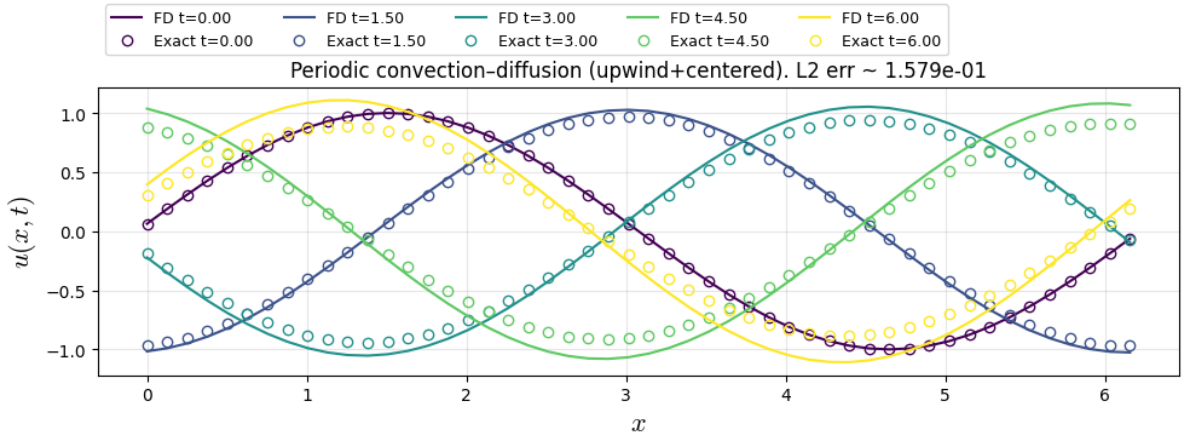
If you use a cosine initial condition $u(x, 0) = A\cos(kx + \phi)$, the solution is the same form: $\$u(x,t) = Ae^{-\nu k^2 t}\cos\left(k(x - ct) + \phi\right).\$$

4.2.1 LBM D1Q3 Solution

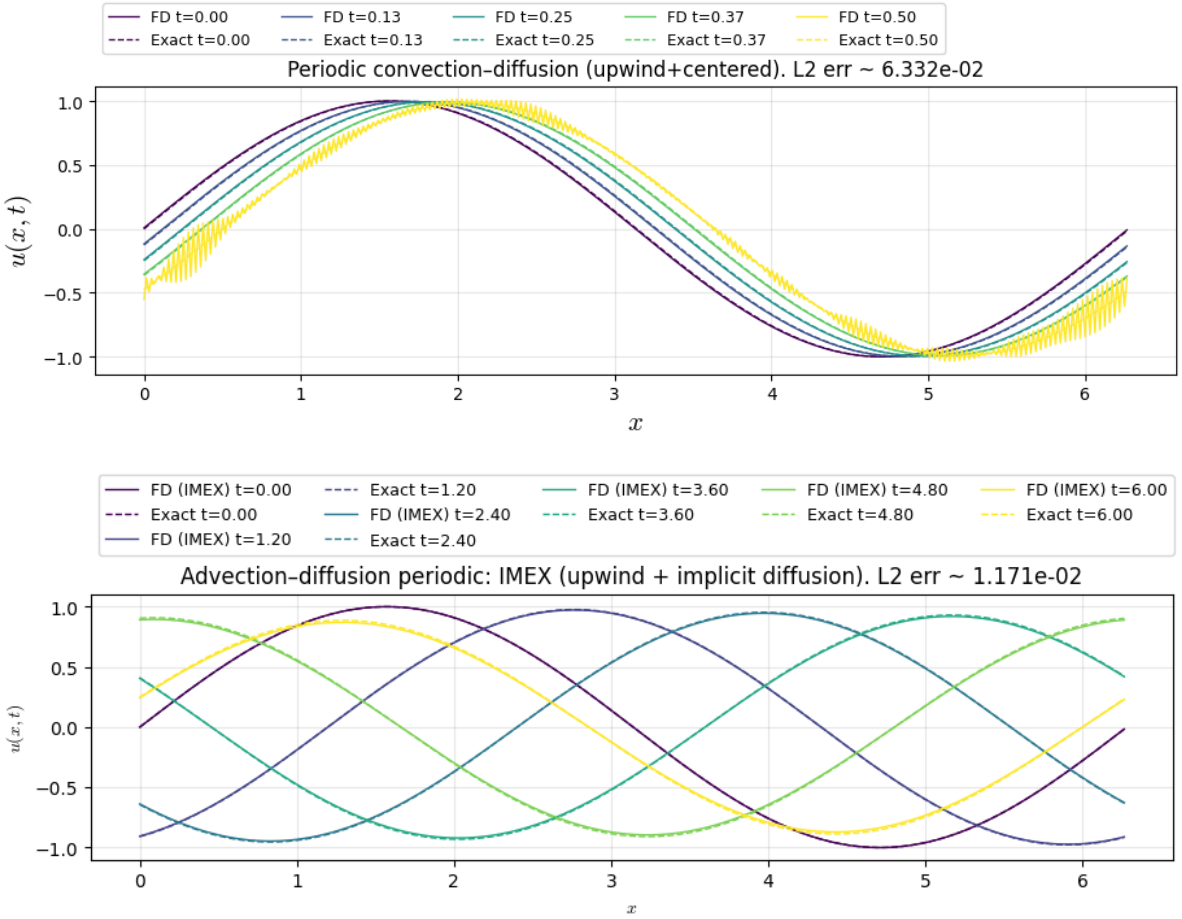
4.2.2 FDM Solution (FTCS-SD) (LBM Framework $\tau = 1$)



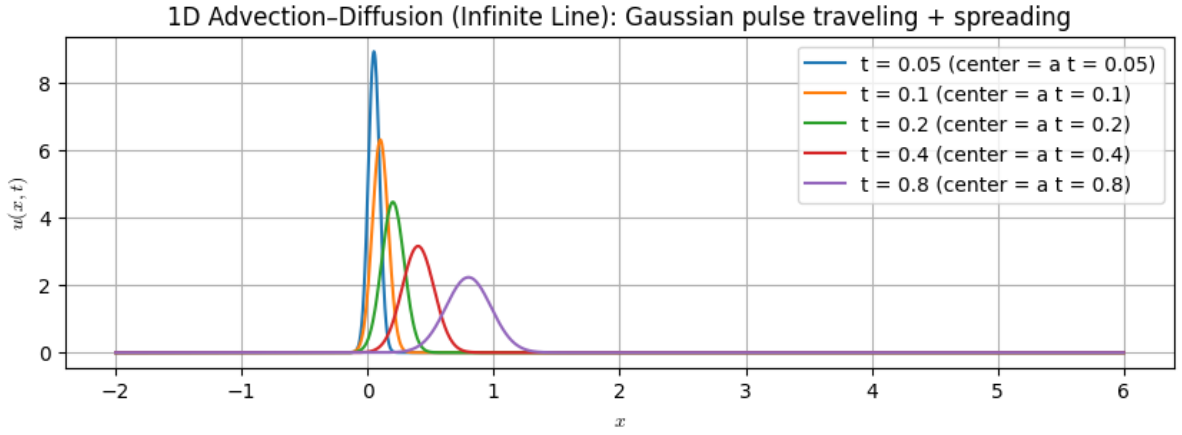
4.2.3 FDM Solution (FTCS)

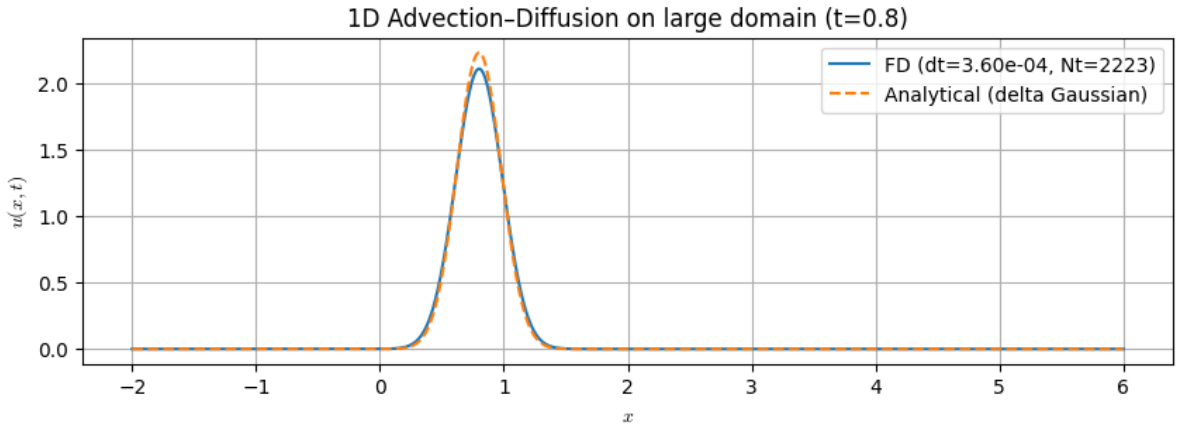


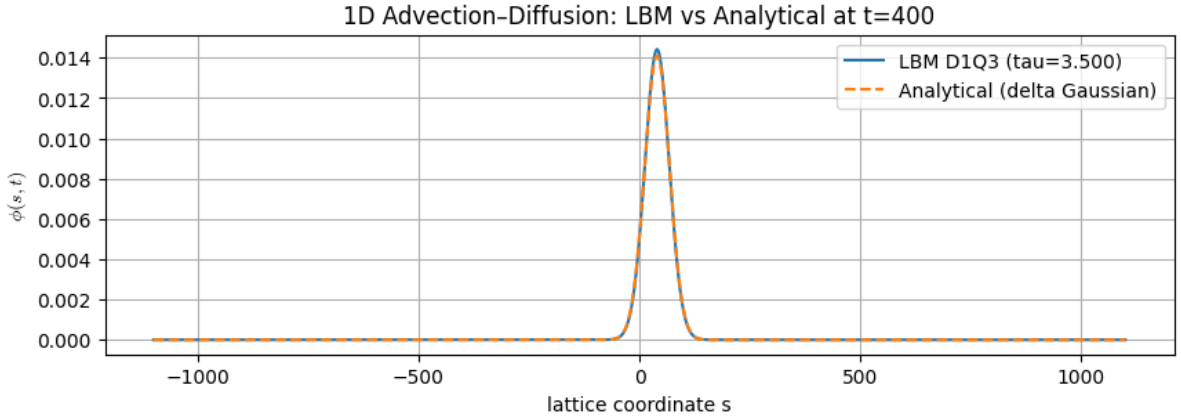
4.2.4 FDM Solution (FTFSCS)



4.3 2nd Benchmark: 1D Perturbation wave-decay




1D Advection–Diffusion on large domain (t=0.8)
FD (dt=3.60e-04, Nt=2223)
Analytical (delta Gaussian)
$u(x,t)$
x


1D Advection–Diffusion: LBM vs Analytical at t=400
LBM D1Q3 (tau=3.500)
Analytical (delta Gaussian)
$\phi(s,t)$
lattice coordinate s

CHAPTER

FIVE

NAVIER-STOKES EQUATION

The Navier-Stokes equation is a non-linear PDE able to describe mass and momentum balance of fluid flow problems, commonly written as

$$\underbrace{\partial_t \rho + \partial_\alpha \rho u_\alpha = 0}_{\text{Mass Balance Equation}},$$

$$\underbrace{\partial_t \rho u_\alpha + \partial_\beta \left(\rho u_\alpha \rho u_\beta\right) = \partial_\beta \left(-p\delta_{\alpha\beta} + \mu\left(\partial_\alpha u_\beta + \partial_\beta u_\alpha\right)\right) + \rho g_\alpha}_{\text{Momentum Balance Equation}}, \tag{5.1}$$

where ρ is the fluid density, u_α is the α-th component of the velocity vector field, p is the pressure field, $\delta_{\alpha\beta}$ is the Kronecker delta, μ is the dynamic viscosity (a constitutive parameter controlling the intensity of viscous diffusion), and g_α is the α-th component of the external body acceleration (e.g., gravity).

5.1 Numerical Discretization: Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation:

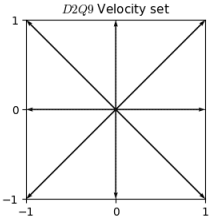
$$\underbrace{f_i\left(x_\alpha + c_{i,\alpha}\Delta t, t + \Delta t\right) - f_i(x_\alpha, t) = -\left(\frac{f_i - f_i^{eq}}{\tau}\right)}_{\text{1st order space-time discretization}}. \tag{5.2}$$

$$\underbrace{\hat{f}_i\left(x_\alpha + c_{i,\alpha}\Delta t, t + \Delta t\right) - \hat{f}_i(x_\alpha, t) = -\left(\frac{\hat{f}_i - \hat{f}_i^{eq}}{\tau}\right) + \left(1 - \frac{1}{2\tau}F_i\right)}_{\text{2nd order space-time discretization}}. \tag{5.3}$$

The equilibrium distribution function is defined by

$$f_i^{eq} = w_i \rho \left(1 + \frac{c_{i,\alpha}u_\alpha}{c_s^2} + \frac{u_\alpha u_\beta \left(c_{i,\alpha}c_{i,\beta} - c_s^2\delta_{\alpha\beta}\right)}{2c_s^4}\right).$$

5.1.1 Lattice Direction Moments: D2Q9 Velocity-Space Discetization



D2Q9 lattice properties

i	c_x	c_y	**Directions**	$\|c\|$	w_i **(exact)**	w_i **(float)**
0	0	0	rest	0 (rest)	$\frac{4}{9}$	0.444444
1	1	0	East	1 (face)	$\frac{1}{9}$	0.111111
2	-1	0	West	1 (face)	$\frac{1}{9}$	0.111111
3	0	1	North	1 (face)	$\frac{1}{9}$	0.111111
4	0	-1	South	1 (face)	$\frac{1}{9}$	0.111111
5	1	1	NE	$\sqrt{2}$ (Diag)	$\frac{1}{36}$	0.027778
6	-1	1	NW	$\sqrt{2}$ (Diag)	$\frac{1}{36}$	0.027778
7	-1	-1	SW	$\sqrt{2}$ (Diag)	$\frac{1}{36}$	0.027778
8	1	-1	SE	$\sqrt{2}$ (Diag)	$\frac{1}{36}$	0.027778

Hermite hierarchy

$\mathcal{H}^0$

$\mathcal{H}_x$ $\mathcal{H}_y$

$\mathcal{H}_{xx}$ $\mathcal{H}_{xy}$ $\mathcal{H}_{yy}$

$\mathcal{H}_{xxy}$ $\mathcal{H}_{xyy}$

$\mathcal{H}_{xxyy}$

Raw Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Moment}} = \rho, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} = \rho u_x, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Moment}} = \rho u_y$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} = \rho \left(u_x^2 + \frac{1}{3} \right), \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} = \rho u_x u_y, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} = \rho \left(u_y^2 + \frac{1}{3} \right)$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} = \rho u_x, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xxy-Third-Order Moment}} = \frac{\rho u_y}{3}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} = \frac{\rho u_x}{3}, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} = \rho u_y$$

Hermite Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Hermite Moment}} = \rho, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} = \rho u_x, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Hermite Moment}} = \rho u_y$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} = \rho u_x^2, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} = \rho u_x u_y, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} = \rho u_y^2$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} = 0, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xxy-Third-Order Moment}} = 0, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} = 0, \qquad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} = 0$$

5.1.2 Chapmann-Enskog Analysis: 1st order Lattice Boltzmann Equation

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i \left(x + c_{i,\alpha} \Delta t, t + \Delta t \right)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i \left(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t \right) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = - \left(\frac{f_i - f_i^{eq}}{\tau} \right), \tag{5.4}$$

where $D_t f_i = \partial_t f_i + c_{i,\alpha} \partial_\alpha f_i$.

Rescaling to the dimensionless form of the Eq. (5.4), we rewrite it in terms of the Knudsen number (Kn) following the procedure presented in Kremer [Kre10] Section 3.3.1:

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = - \frac{1}{Kn} \left(\frac{f_i - f_{eq,i}}{\tau} \right) \qquad \rightarrow \qquad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = - \frac{f_i - f_i^{eq}}{\tau},$$

expanding the above equation up to fourth-order:

$$
\begin{aligned}
\frac{f_i^{eq} - f_i}{\tau} = & Kn^{(1)} \left(c_{i,\alpha} \partial_\alpha f_i + \partial_t f_i \right) \\
& + Kn^{(2)} \left(\frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i}{2} + c_{i,\alpha} \partial_\alpha \partial_t f_i + \frac{\partial_t^2 f_i}{2} \right) \\
& + Kn^{(3)} \left(\frac{c_{i,\alpha}^3 \partial_\alpha^3 f_i}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t f_i}{2} + \frac{c_{i,\alpha} \partial_\alpha \partial_t^2 f_i}{2} + \frac{\partial_t^3 f_i}{6} \right) \\
& + Kn^{(4)} \left(\frac{c_{i,\alpha}^4 \partial_\alpha^4 f_i}{24} + \frac{c_{i,\alpha}^3 \partial_\alpha^3 \partial_t f_i}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^2 f_i}{4} + \frac{c_{i,\alpha} \partial_\alpha \partial_t^3 f_i}{6} + \frac{\partial_t^4 f_i}{24} \right)
\end{aligned}
$$

Applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \cdots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn \partial_t^{(1)} + Kn^2 \partial_t^{(2)} + \cdots$):

$$
\begin{aligned}
& - \frac{f_i^{(0)}}{\tau} + \frac{f_i^{eq}}{\tau} + - \frac{Kn f_i^{(1)}}{\tau} + - \frac{Kn^2 f_i^{(2)}}{\tau} + - \frac{Kn^3 f_i^{(3)}}{\tau} + - \frac{Kn^4 f_i^{(4)}}{\tau} \quad = \\
& Kn \left(c_{i,\alpha} \partial_\alpha f_i^{(0)} + \partial_t^{(0)} f_i^{(0)} \right) \\
& + Kn^2 \left(\frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i^{(0)}}{2} + c_{i,\alpha} \partial_\alpha f_i^{(1)} + c_{i,\alpha} \partial_\alpha \partial_t^{(0)} f_i^{(0)} + \frac{\left(\partial_t^{(0)} \right)^2 f_i^{(0)}}{2} + \partial_t^{(1)} f_i^{(0)} + \partial_t^{(0)} f_i^{(1)} \right) \\
& + Kn^3 \left(\frac{c_{i,\alpha}^3 \partial_\alpha^3 f_i^{(0)}}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i^{(1)}}{2} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^{(0)} f_i^{(0)}}{2} + c_{i,\alpha} \partial_\alpha f_i^{(2)} + \frac{c_{i,\alpha} \partial_\alpha \left(\partial_t^{(0)} \right)^2 f_i^{(0)}}{2} + c_{i,\alpha} \partial_\alpha \partial_t^{(1)} f_i^{(0)} + c_{i,\alpha} \partial_\alpha \partial_t^{(0)} f_i^{(1)} + \frac{\left(\partial_t^{(0)} \right)^3 f_i^{(0)}}{6} \right. \\
& + Kn^4 \left(\frac{c_{i,\alpha}^4 \partial_\alpha^4 f_i^{(0)}}{24} + \frac{c_{i,\alpha}^3 \partial_\alpha^3 f_i^{(1)}}{6} + \frac{c_{i,\alpha}^3 \partial_\alpha^3 \partial_t^{(0)} f_i^{(0)}}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i^{(2)}}{2} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \left(\partial_t^{(0)} \right)^2 f_i^{(0)}}{4} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^{(1)} f_i^{(0)}}{2} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^{(0)} f_i^{(1)}}{2} + c_{i,\alpha} \partial_\alpha f_i^{(3)} \right.
\end{aligned}
$$

Splitting the equation in orders up to the order Kn^3 (References with High-Order expansion in Knudsen-order: Chai *et al.* [CHGS18]):

$$
\begin{aligned}
& (Kn^{(0)}): \quad f_i^{(0)} = f_i^{eq}, \\
& (Kn^{(1)}): \quad \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) f_i^{(0)} = - \frac{f_i^{(1)}}{\tau}, \\
& (Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) f_i^{(1)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right)^2 f_i^{(0)}}{2} = - \frac{f_i^{(2)}}{\tau}, \\
& \text{ou} \\
& (Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) \left(1 - \frac{1}{2\tau} \right) f_i^{(1)} = - \frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(1)} \text{ Formulation} \\
& \text{ou} \\
& (Kn^{(2)}): \quad \partial_t^{(1)} f_i^{(0)} + \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right)^2 \left(\frac{1}{2} - \tau \right) f_i^{(0)} = - \frac{f_i^{(2)}}{\tau} \qquad \rightarrow \qquad f_i^{(0)} \text{ Formulation} \\
& (Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)} \left(\left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) f_i^{(0)} \right) + \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) f_i^{(2)} + \frac{\left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right)^2 f_i^{(1)}}{2} \\
& \qquad + \frac{\left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right)^3 f_i^{(0)}}{6} = - \frac{f_i^{(3)}}{\tau}, \\
& \text{ou} \\
& (Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + \left(2 - \frac{1}{\tau} \right) \partial_t^{(1)} f_i^{(1)} + \left(1 - \frac{1}{6\tau} - \tau \right) \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right)^2 f_i^{(1)} = - \frac{f_i^{(3)}}{\tau}, \qquad \rightarrow \qquad f_i^{(1)} \text{ Formulation} \\
& \text{ou} \\
& (Kn^{(3)}): \quad \partial_t^{(2)} f_i^{(0)} + 2 \left(\frac{1}{2} - \tau \right) \partial_t^{(1)} \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) f_i^{(0)} + \left(\tau^2 - \tau + \frac{1}{6} \right) \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right)^3 f_i^{(0)} = - \frac{f_i^{(3)}}{\tau}, \qquad \rightarrow \qquad f_i^{(0)} \text{ Formulation}
\end{aligned}
\tag{5.5}
$$

Zero-Order Moment Balance

To retrieve the mass balance equation, we sum the Eq. (5.5) for $Kn^{(1)}$, $Kn^{(2)}$, and $Kn^{(3)}$ over $\sum_{i=0}$:

$$\begin{aligned}
(Kn^{(1)}): &\quad \sum_i \left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(0)} = -\underbrace{\sum_i \left(\frac{f_i^{(1)}}{\tau}\right)}_{=0}, \\
(Kn^{(1)}): &\quad \partial_t^{(0)}\rho + \partial_\alpha\left(\rho u_\alpha\right) = 0,
\end{aligned} \tag{5.6}$$

$$\begin{aligned}
(Kn^{(2)}): &\quad \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1-\frac{1}{2\tau}\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right) f_i^{(1)}\right] = -\sum_i \left(\frac{f_i^{(2)}}{\tau}\right), \\
(Kn^{(2)}): &\quad \partial_t^{(1)}\sum_i f_i^{(0)} + \underbrace{\left(1-\frac{1}{2\tau}\right)\partial_t^{(0)}\sum_i f_i^{(1)}}_{=0} + \underbrace{\left(1-\frac{1}{2\tau}\right)\partial_\alpha\sum_i c_{i,\alpha} f_i^{(1)}}_{=0} = \underbrace{-\frac{1}{\tau}\sum_i f_i^{(2)}}_{=0} \\
(Kn^{(2)}): &\quad \partial_t^{(1)}\rho = 0,
\end{aligned}$$

Extending the analysis up to $Kn^{(3)}$

$$\begin{aligned}
(Kn^{(3)}): &\quad \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(2-\frac{1}{\tau}\right)\partial_t^{(1)} f_i^{(1)} + \left(1-\frac{1}{6\tau}-\tau\right)\left(\partial_t^{(0)} + c_{i,\alpha}\partial_\alpha\right)^2 f_i^{(1)}\right] = -\sum_i \left[\frac{f_i^{(3)}}{\tau}\right], \\
(Kn^{(3)}): &\quad \partial_t^{(2)}\sum_i f_i^{(0)} + \underbrace{\left(2-\frac{1}{\tau}\right)\partial_t^{(1)}\sum_i f_i^{(1)}}_{=0} + \left(1-\frac{1}{6\tau}-\tau\right)\left(\underbrace{\partial_t^{(0)}\partial_t^{(0)}\sum_i f_i^{(1)}}_{=0} + \underbrace{\partial_\alpha\partial_t^{(0)}\sum_i c_{i,\alpha} f_i^{(1)}}_{=0} + \underbrace{\partial_\beta\partial_t^{(0)}\sum_i c_{i,\beta} f_i^{(1)}}_{=0}\right) \\
&\quad + \left(1-\frac{1}{6\tau}-\tau\right)\left(\partial_\alpha\partial_\beta\sum_i c_{i,\alpha}c_{i,\beta} f_i^{(1)}\right) = -\frac{1}{\tau}\underbrace{\sum_i f_i^{(3)}}_{=0}, \\
(Kn^{(3)}): &\quad \partial_t^{(2)}\rho + \left(1-\frac{1}{6\tau}-\tau\right)\partial_\alpha\partial_\beta m_{\alpha\beta}^{(1)} = 0.
\end{aligned} \tag{5.7}$$

First-Order Moment Balance

To retrieve the momentum balance equation, we sum the Eq. (5.5) for $Kn^{(1)}$, $Kn^{(2)}$, and $Kn^{(3)}$ over $\sum_{i=0} c_{i,\alpha}$:

$$\begin{aligned}
(Kn^{(1)}): &\sum_i \left(\partial_t^{(0)} + c_{i,\beta}\partial_\beta\right) f_i^{(0)} c_{i,\alpha} = -\underbrace{\sum_i \left(\frac{f_i^{(1)}}{\tau}\right) c_{i,\alpha}}_{=0}, \\
(Kn^{(1)}): &\partial_t^{(0)}\left(\rho u_\alpha\right) + \partial_\beta\left(\rho u_\alpha u_\beta + \rho c_s^2\delta_{\alpha\beta}\right) = 0,
\end{aligned} \tag{5.8}$$

$$\begin{aligned}
(Kn^{(2)}): &\quad \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1-\frac{1}{2\tau}\right)\left(\partial_t^{(0)} + c_{i,\beta}\partial_\beta\right) f_i^{(1)}\right] c_{i,\alpha} = -\sum_i \left(\frac{f_i^{(2)}}{\tau}\right) c_{i,\alpha}, \\
(Kn^{(2)}): &\quad \partial_t^{(1)}\sum_i f_i^{(0)} c_{i,\alpha} + \underbrace{\left(1-\frac{1}{2\tau}\right)\partial_t^{(0)}\sum_i f_i^{(1)} c_{i,\alpha}}_{=0} + \left(1-\frac{1}{2\tau}\right)\partial_\beta\sum_i f_i^{(1)} c_{i,\alpha}c_{i,\beta} = \underbrace{-\frac{1}{\tau}\sum_i f_i^{(2)} c_{i,\alpha}}_{=0} \\
(Kn^{(2)}): &\quad \partial_t^{(1)}\left(\rho u_\alpha\right) + \left(1-\frac{1}{2\tau}\right)\partial_\beta m_{\alpha\beta}^{(1)} = 0,
\end{aligned}$$

Extending the analysis up to $Kn^{(3)}$

$$
\begin{aligned}
(Kn^{(3)}): \quad & \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(2-\frac{1}{\tau}\right)\partial_t^{(1)} f_i^{(1)} + \left(1-\frac{1}{6\tau}-\tau\right)\left(\partial_t^{(0)} + c_{i,\beta}\partial_\beta\right)^2 f_i^{(1)}\right] c_{i,\alpha} = -\sum_i \left[\frac{f_i^{(3)}}{\tau}\right] c_{i,\alpha}, \\
(Kn^{(3)}): \quad & \partial_t^{(2)} \sum_i f_i^{(0)} c_{i,\alpha} + \underbrace{\left(2-\frac{1}{\tau}\right)\partial_t^{(1)} \sum_i f_i^{(1)} c_{i,\alpha}}_{=0} + \left(1-\frac{1}{6\tau}-\tau\right)\left(\underbrace{\partial_t^{(0)}\partial_t^{(0)} \sum_i f_i^{(1)} c_{i,\alpha}}_{=0}\right) \\
& + \left(1-\frac{1}{6\tau}-\tau\right)\left(\partial_t^{(0)}\partial_\beta \sum_i f_i^{(1)} c_{i,\alpha} c_{i,\beta} + \partial_t^{(0)}\partial_\gamma \sum_i f_i^{(1)} c_{i,\alpha} c_{i,\gamma} + \partial_\beta\partial_\gamma \sum_i c_{i,\alpha} c_{i,\beta} c_{i,\gamma} f_i^{(1)}\right) = -\frac{1}{\tau}\underbrace{\sum_i f_i^{(3)}}_{=0}, \\
(Kn^{(3)}): \quad & \partial_t^{(2)}\left(\rho u_\alpha\right) + \left(1-\frac{1}{6\tau}-\tau\right)\left(\partial_t^{(0)}\partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)}\partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta\partial_\gamma m_{\alpha\beta\gamma}^{(1)}\right) = 0.
\end{aligned}
\tag{5.9}
$$

Second-Order Non-Equilibrium Moment

To determine the unknow term $m_{\alpha\beta}^{(1)}$, we sum the Eq. (5.5) for $Kn^{(1)}$ over the second-order moment:

$$
\begin{aligned}
(Kn^{(1)}): & \sum_i \left(\partial_t^{(0)} + c_{i,\gamma}\partial_\gamma\right) f_i^{(0)} c_{i,\alpha} c_{i,\beta} = -\sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha} c_{i,\beta}\right), \\
(Kn^{(1)}): & \partial_t^{(0)}\left(\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}\right) + \partial_\gamma\left(c_s^2\rho\left(u_\alpha\delta_{\beta\gamma} + u_\beta\delta_{\alpha\gamma} + u_\gamma\delta_{\alpha\beta}\right)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau},
\end{aligned}
$$

notice that third-order moment for the lattice D2Q9 is not correct recovered. Expanding the terms and using Eqs. (5.6) and (5.8) for numerical manipulation:

$$
\begin{aligned}
(Kn^{(1)}): \quad & \partial_t^{(0)}\left(\rho u_\alpha u_\beta + \rho c_s^2\delta_{\alpha\beta}\right) + \partial_\gamma\left(c_s^2\rho\left(u_\alpha\delta_{\beta\gamma} + u_\beta\delta_{\alpha\gamma} + u_\gamma\delta_{\alpha\beta}\right)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & \partial_t^{(0)}\left(\rho c_s^2\delta_{\alpha\beta}\right) + \partial_t^{(0)}\left(\rho u_\alpha u_\beta\right) + c_s^2\left(\partial_\gamma(\rho u_\alpha)\delta_{\beta\gamma} + \partial_\gamma(\rho u_\beta)\delta_{\alpha\gamma}\right) + \partial_\gamma\left(c_s^2\rho u_\gamma\delta_{\alpha\beta}\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & c_s^2\partial_t^{(0)}\left(\rho\right)\delta_{\alpha\beta} + \rho u_\alpha\partial_t^{(0)}\left(u_\beta\right) + u_\beta\partial_t^{(0)}\left(\rho u_\alpha\right) + c_s^2\left(\partial_\beta(\rho u_\alpha) + \partial_\alpha(\rho u_\beta)\right) + c_s^2\partial_\gamma\left(\rho u_\gamma\right)\delta_{\alpha\beta} = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & -c_s^2\partial_\gamma\left(\rho u_\gamma\right)\delta_{\alpha\beta} + \rho u_\alpha\partial_t^{(0)}\left(u_\beta\right) - u_\beta\partial_\gamma\left(\rho u_\alpha u_\gamma + \rho c_s^2\delta_{\alpha\gamma}\right) + c_s^2\left(\partial_\beta(\rho u_\alpha) + \partial_\alpha(\rho u_\beta)\right) + c_s^2\partial_\gamma\left(\rho u_\gamma\right)\delta_{\alpha\beta} = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & u_\alpha\left(-u_\beta\partial_t^{(0)}\rho - \partial_\gamma\left(\rho u_\beta u_\gamma + \rho c_s^2\delta_{\beta\gamma}\right)\right) - u_\beta\partial_\gamma\left(\rho u_\alpha u_\gamma + \rho c_s^2\delta_{\alpha\gamma}\right) + c_s^2\left(\partial_\beta(\rho u_\alpha) + \partial_\alpha(\rho u_\beta)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & u_\alpha\left(u_\beta\partial_\gamma(\rho u_\gamma) - \partial_\gamma\left(\rho u_\beta u_\gamma + \rho c_s^2\delta_{\beta\gamma}\right)\right) - u_\beta\partial_\gamma\left(\rho u_\alpha u_\gamma + \rho c_s^2\delta_{\alpha\gamma}\right) + c_s^2\left(\partial_\beta(\rho u_\alpha) + \partial_\alpha(\rho u_\beta)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & u_\alpha\left(-\rho u_\gamma\partial_\gamma u_\beta - c_s^2\partial_\beta\rho\right) - u_\beta\partial_\gamma\left(\rho u_\alpha u_\gamma + \rho c_s^2\delta_{\alpha\gamma}\right) + c_s^2\left(\partial_\beta(\rho u_\alpha) + \partial_\alpha(\rho u_\beta)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & -\rho u_\alpha u_\gamma\partial_\gamma u_\beta - c_s^2 u_\alpha\partial_\beta\rho - u_\beta\partial_\gamma\rho u_\alpha u_\gamma - u_\beta c_s^2\partial_\alpha\rho + c_s^2\left(\partial_\beta(\rho u_\alpha) + \partial_\alpha(\rho u_\beta)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & -\partial_\gamma(\rho u_\alpha u_\beta u_\gamma) - c_s^2(u_\alpha\partial_\beta\rho + u_\beta\partial_\alpha\rho) + c_s^2\left(\partial_\beta(\rho u_\alpha) + \partial_\alpha(\rho u_\beta)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & -\partial_\gamma(\rho u_\alpha u_\beta u_\gamma) + c_s^2\rho\left(\partial_\beta u_\alpha + \partial_\alpha u_\beta\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \qquad \rightarrow \qquad m_{\alpha\beta}^{(1)} = -c_s^2\tau\rho\left(\partial_\beta u_\alpha + \partial_\alpha u_\beta\right) + \tau\partial_\gamma(\rho u_\alpha u_\beta u_\gamma)
\end{aligned}
\tag{5.10}
$$

Balance Equations

To recover the mass balance equation, we sum the Eqs. (5.6), (5.7), and (5.7):

$$\partial_t \rho + \partial_\alpha(\rho u_\alpha) + \underbrace{\left(1 - \frac{1}{2\tau}\right)\partial_\beta m^{(1)}_{\alpha\beta} + ... + \mathcal{O}(Kn^{(\geq 3)})}_{\text{Numerical Errors}} = 0,$$

$$\partial_t \rho + \partial_\alpha(\rho u_\alpha) + \underbrace{\left(\tau - \frac{1}{2}\right)\left(\partial_\beta(-c_s^2\rho\left(\partial_\beta u_\alpha + \partial_\alpha u_\beta\right) + \partial_\gamma(\rho u_\alpha u_\beta u_\gamma))\right) + ... + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0,$$

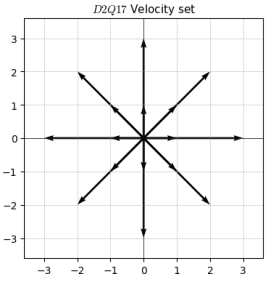
Repeting the process for momentum balance, we sum the Eqs. (5.8), (5.9), (5.9), and (5.10):

$$\partial_t \rho u_\alpha + \partial_\beta\left(\rho u_\alpha \rho u_\beta\right) = \partial_\beta\left(-\rho c_s^2 \delta_{\alpha\beta} + \left(\tau - \frac{1}{2}\right)\rho c_s^2\left(\partial_\alpha u_\beta + \partial_\beta u_\alpha\right)\right)$$
$$\underbrace{-\left(\tau - \frac{1}{2}\right)\partial_\gamma(\rho u_\alpha u_\beta u_\gamma) - \left(1 - \frac{1}{6\tau} - \tau\right)\left(\partial_t^{(0)}\partial_\beta m^{(1)}_{\alpha\beta} + \partial_t^{(0)}\partial_\gamma m^{(1)}_{\alpha\gamma} + \partial_\beta\partial_\gamma m^{(1)}_{\alpha\beta\gamma}\right) + ... + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0,$$

$$\partial_t \rho u_\alpha + \partial_\beta\left(\rho u_\alpha \rho u_\beta\right) = \partial_\beta\left(-p\delta_{\alpha\beta} + \mu\left(\partial_\alpha u_\beta + \partial_\beta u_\alpha\right)\right)$$
$$\underbrace{-\left(\tau - \frac{1}{2}\right)\partial_\gamma(\rho u_\alpha u_\beta u_\gamma) - \left(1 - \frac{1}{6\tau} - \tau\right)\left(\partial_t^{(0)}\partial_\beta m^{(1)}_{\alpha\beta} + \partial_t^{(0)}\partial_\gamma m^{(1)}_{\alpha\gamma} + \partial_\beta\partial_\gamma m^{(1)}_{\alpha\beta\gamma}\right) + ... + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0,$$

where $\mu = \rho c_s^2(\tau - 1/2)$.

5.1.3 Lattice Direction Moments: (Multi-Speed Lattices) D2V17 Velocity-Space Discetization



D2Q17 lattice properties

i	c_x	c_y	Directions	$\lvert c_i \rvert$	w_i (exact)	w_i (float)
0	0	0	rest	0 (rest)	$\frac{36}{575} + \frac{19\sqrt{70}}{1150}$	0.402005
1	1	0	East	1 (face)	$\frac{479}{8050} - \frac{9\sqrt{70}}{1610}$	0.116155
2	0	1	North	1 (face)	$\frac{479}{8050} - \frac{9\sqrt{70}}{1610}$	0.116155
3	-1	0	West	1 (face)	$\frac{479}{8050} - \frac{9\sqrt{70}}{1610}$	0.116155
4	0	-1	South	1 (face)	$\frac{479}{8050} - \frac{9\sqrt{70}}{1610}$	0.116155
5	1	1	NE	$\sqrt{2}$ (1st diag)	$\frac{17\sqrt{2}}{2300} + \frac{131}{2300}$	0.033006
6	-1	1	NW	$\sqrt{2}$ (1st diag)	$\frac{17\sqrt{2}}{2300} + \frac{131}{2300}$	0.033006
7	-1	-1	SW	$\sqrt{2}$ (1st diag)	$\frac{17\sqrt{2}}{2300} + \frac{131}{2300}$	0.033006
8	1	-1	SE	$\sqrt{2}$ (1st diag)	$\frac{17\sqrt{2}}{2300} + \frac{131}{2300}$	0.033006
9	2	2	NE$_2$	$2\sqrt{2}$ (2nd diag)	$\frac{47}{59800} - \frac{81\sqrt{2}}{59800}$	0.000079
10	-2	2	NW$_2$	$2\sqrt{2}$ (2nd diag)	$\frac{47}{59800} - \frac{81\sqrt{2}}{59800}$	0.000079
11	-2	-2	SW$_2$	$2\sqrt{2}$ (2nd diag)	$\frac{47}{59800} - \frac{81\sqrt{2}}{59800}$	0.000079
12	2	-2	SE$_2$	$2\sqrt{2}$ (2nd diag)	$\frac{47}{59800} - \frac{81\sqrt{2}}{59800}$	0.000079
13	3	0	E$_3$	3 (3rd face)	$\frac{201}{11400} - \frac{611\sqrt{133}}{912000}$	0.000258
14	0	3	N$_3$	3 (3rd face)	$\frac{201}{11400} - \frac{611\sqrt{133}}{912000}$	0.000258
15	-3	0	W$_3$	3 (3rd face)	$\frac{201}{11400} - \frac{611\sqrt{133}}{912000}$	0.000258
16	0	-3	S$_3$	3 (3rd face)	$\frac{201}{11400} - \frac{611\sqrt{133}}{912000}$	0.000258

Hermite hierarchy

$\mathcal{H}^0$

$\mathcal{H}_x$ $\mathcal{H}_y$

$\mathcal{H}_{xx}$ $\mathcal{H}_{xy}$ $\mathcal{H}_{yy}$

$\mathcal{H}_{xxx}$ $\mathcal{H}_{xxy}$ $\mathcal{H}_{xyy}$ $\mathcal{H}_{yyy}$

$\mathcal{H}_{xxxy}$ $\mathcal{H}_{xxyy}$ $\mathcal{H}_{xyyy}$

$\mathcal{H}_{xxxyy}$ $\mathcal{H}_{xxyyy}$

$\mathcal{H}_{xxxyyy}$

Raw Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i = \rho,}_{\text{Zero-Order Moment}} \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} = \rho u_x^*,}_{\text{x-First-Order Moment}} \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} = \rho u_y^*}_{\text{y-First-Order Moment}}$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} = \rho c_s^2 + \rho\left(u_x^*\right)^2,}_{\text{xx-Second-Order Moment}} \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} = \rho u_x^* u_y^*,}_{\text{xy-Second-Order Moment}} \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} = \rho c_s^2 + \rho\left(u_y^*\right)^2}_{\text{yy-Second-Order Moment}}$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} = 3\rho c_s^2 u_x^* + \rho \left(u_x^*\right)^3, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xxy-Third-Order Moment}} = \rho c_s^2 u_y^* + \rho \left(u_x^*\right)^2 u_y^*,$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} = \rho c_s^2 u_x^* + \rho u_x^* \left(u_y^*\right)^2, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} = 3\rho c_s^2 u_y^* + \rho \left(u_y^*\right)^3$$

Hermite Moments

$$\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Hermite Moment}} = \rho, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} = \rho u_x^*, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Hermite Moment}} = \rho u_y^*$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} = \rho \left(u_x^*\right)^2, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} = \rho u_x^* u_y^*, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} = \rho \left(u_y^*\right)^2$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} = \rho \left(u_x^*\right)^3, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xxy-Third-Order Moment}} = \rho \left(u_x^*\right)^2 u_y^*,$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} = \rho u_x^* \left(u_y^*\right)^2, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} = \rho \left(u_y^*\right)^3$$

Doing the Chapman-Enskog analysis for the lattice D2V17, we obtain the same Eqs. (5.6), (5.7), (5.7), (5.8), (5.9), and (5.9). However, the calculation of the second-order moment of non-equilibrium $m_{\alpha\beta}^{(1)}$ will change due to the D2V17 be able to correctly recover the third-order moments:

$$\begin{aligned}
(Kn^{(1)}): \quad & \sum_i \left(\partial_t^{(0)} + c_{i,\gamma}\partial_\gamma\right) f_i^{(0)} c_{i,\alpha} c_{i,\beta} = -\sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha} c_{i,\beta}\right), \\
(Kn^{(1)}): \quad & \partial_t^{(0)} \left(\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}\right) + \partial_\gamma \left(c_s^2 \rho \left(u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}\right)\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & \partial_t^{(0)} \left(\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}\right) + \partial_\gamma \left(c_s^2 \rho \left(u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}\right)\right) + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & \partial_t^{(0)} \left(\rho c_s^2 \delta_{\alpha\beta}\right) + \partial_t^{(0)} \left(\rho u_\alpha u_\beta\right) + c_s^2 \left(\partial_\gamma (\rho u_\alpha) \delta_{\beta\gamma} + \partial_\gamma (\rho u_\beta) \delta_{\alpha\gamma}\right) + \partial_\gamma \left(c_s^2 \rho u_\gamma \delta_{\alpha\beta}\right) + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & c_s^2 \partial_t^{(0)} \left(\rho\right) \delta_{\alpha\beta} + \rho u_\alpha \partial_t^{(0)} \left(u_\beta\right) + u_\beta \partial_t^{(0)} \left(\rho u_\alpha\right) + c_s^2 \left(\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)\right) + c_s^2 \partial_\gamma \left(\rho u_\gamma\right) \delta_{\alpha\beta} + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & -c_s^2 \partial_\gamma \left(\rho u_\gamma\right) \delta_{\alpha\beta} + \rho u_\alpha \partial_t^{(0)} \left(u_\beta\right) - u_\beta \partial_\gamma \left(\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}\right) + c_s^2 \left(\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)\right) + c_s^2 \partial_\gamma \left(\rho u_\gamma\right) \delta_{\alpha\beta} + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & u_\alpha \left(-u_\beta \partial_t^{(0)} \rho - \partial_\gamma \left(\rho u_\beta u_\gamma + \rho c_s^2 \delta_{\beta\gamma}\right)\right) - u_\beta \partial_\gamma \left(\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}\right) + c_s^2 \left(\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)\right) + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & u_\alpha \left(u_\beta \partial_\gamma (\rho u_\gamma) - \partial_\gamma \left(\rho u_\beta u_\gamma + \rho c_s^2 \delta_{\beta\gamma}\right)\right) - u_\beta \partial_\gamma \left(\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}\right) + c_s^2 \left(\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)\right) + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & u_\alpha \left(-\rho u_\gamma \partial_\gamma u_\beta - c_s^2 \partial_\beta \rho\right) - u_\beta \partial_\gamma \left(\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}\right) + c_s^2 \left(\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)\right) + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & -\rho u_\alpha u_\gamma \partial_\gamma u_\beta - c_s^2 u_\alpha \partial_\beta \rho - u_\beta \partial_\gamma \rho u_\alpha u_\gamma - u_\beta c_s^2 \partial_\alpha \rho + c_s^2 \left(\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)\right) + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & -\partial_\gamma (\rho u_\alpha u_\beta u_\gamma) - c_s^2 (u_\alpha \partial_\beta \rho + u_\beta \partial_\alpha \rho) + c_s^2 \left(\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)\right) + \partial_\gamma \left(u_\alpha u_\beta u_\gamma\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
(Kn^{(1)}): \quad & c_s^2 \rho \left(\partial_\beta u_\alpha + \partial_\alpha u_\beta\right) = -\frac{m_{\alpha\beta}^{(1)}}{\tau}, \qquad \rightarrow \qquad m_{\alpha\beta}^{(1)} = -c_s^2 \tau \rho \left(\partial_\beta u_\alpha + \partial_\alpha u_\beta\right)
\end{aligned} \tag{5.11}$$

Repeting the process for momentum balance, we sum the Eqs. (5.8), (5.9), (5.9), and (5.11):

$$\begin{aligned}
&\partial_t \rho u_\alpha + \partial_\beta \left(\rho u_\alpha \rho u_\beta\right) = \partial_\beta \left(-\rho c_s^2 \delta_{\alpha\beta} + \left(\tau - \frac{1}{2}\right) \rho c_s^2 \left(\partial_\alpha u_\beta + \partial_\beta u_\alpha\right)\right) \\
&\underbrace{- \left(1 - \frac{1}{6\tau} - \tau\right) \left(\partial_t^{(0)} \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)} \partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta \partial_\gamma m_{\alpha\beta\gamma}^{(1)}\right) + ... + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0, \\
&\partial_t \rho u_\alpha + \partial_\beta \left(\rho u_\alpha \rho u_\beta\right) = \partial_\beta \left(-p \delta_{\alpha\beta} + \mu \left(\partial_\alpha u_\beta + \partial_\beta u_\alpha\right)\right) \\
&\underbrace{- \left(1 - \frac{1}{6\tau} - \tau\right) \left(\partial_t^{(0)} \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)} \partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta \partial_\gamma m_{\alpha\beta\gamma}^{(1)}\right) + ... + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0,
\end{aligned}$$

where $\mu = \rho c_s^2 (\tau - 1/2)$.

5.2 Tutorials

Part II

Finite Difference Structure of Lattice Boltzmann Method

CHAPTER

SIX

INTODUCTION TO DIFFERENCE FINITE SCHEME DERIVED FROM LATTICE BOLTZMANN METHOD: DIFFUSIVE EQUATION

Due to the similarities in the grid structures employed by the lattice Boltzmann method and finite-difference schemes, several studies have investigated the direct relationships between these approaches and their mathematical interpretations [Anc94, BGM22, CCS23, DLOT17, Jun01, Sug10]. In this section, through mathematical manipulation of the lattice Boltzmann equations with single relaxation time, we derive a finite-difference interpretation for one of the simplest transport problems, involving purely diffusive transport of a scalar quantity. The present lattice Boltzmann formulation follows the analysis presented by Suga [Sug10].

6.1 Diffusive Equation Interpretation with Single Relaxation Time

Given the LBM formulation for the diffusive equation:

$$f_i(x_\alpha + e_{i,\alpha}\delta t, t+\delta t) = f_i(x_\alpha, t) - \left(\frac{f_i - f_i^{eq}}{\tau}\right) \qquad \rightarrow \qquad f_i(x_\alpha + e_{i,\alpha}\delta t, t+\delta t) = f_i(x_\alpha, t) - \omega\left(f_i - f_i^{eq}\right), \tag{6.1}$$

where the equilibrium distribution function is defined by

$$f_i^{eq} = w_i \phi(x_\alpha, t), \tag{6.2}$$

and the equilibrium moments are given by

$$\sum_{i=0} f_i^{eq} = \phi, \qquad \sum_i f_i^{eq} e_{i,\alpha} = 0 \qquad \text{and} \qquad \sum_i f_i^{eq} e_{i,\alpha} e_{i,\beta} = c_s^2 \phi \delta_{\alpha\beta}, \tag{6.3}$$

where $\phi(x_\alpha, t)$ is the scalar term transported by diffusion.

6.1.1 FDM Interpretation for the Lattice D1Q3

For simplification we redefine the notation for $f_i(x_\alpha + e_{i,\alpha}\delta t, t+\delta t) = f_{i,x+c_i}^{t+1}$. Using this notation, Eq. (6.1) can be reinterpreted by applying a shift in c_i to express the distribution function at the pre-streaming state:

$$f_{i,x}^{t+1} = f_{i,x-c_i}^{t} - \omega\left(f_{i,x-c_i}^{t} - f_{i,x-c_i}^{eq,t}\right). \tag{6.4}$$

Summing the Eq. (6.4) over all lattice direction and replacing f_i^{eq} by Eq. (6.2), we have:

$$\sum_i \left(f_{i,x}^{t+1}\right) = \sum_i \left(f_{i,x-c_i}^{t} - \omega\left(f_{i,x-c_i}^{t} - f_{i,x-c_i}^{eq,t}\right)\right) \qquad \rightarrow \qquad \phi_x^{t+1} = \sum_i \left((1-\omega) f_{i,x-c_i}^{t} + \omega w_i \phi_{x-c_i}^{t}\right),$$

using the Eq. (6.3) to replace $f^t_{i,x-c_i}$ in the above equation:

$$\phi^{t+1}_x = \sum_i \Big((1-\omega) f^t_{i,x-c_i} + \omega w_i \phi^t_{x-c_i} \Big) \qquad \rightarrow \qquad \phi^{t+1}_x = \sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} - (1-\omega)\sum_{k\neq i} f^t_{k,x-c_i} \Big).$$

Spliting the terms in the above equation and using Eq. (6.4) for $f^t_{i,x-c_i}$:

$$\begin{aligned}
\phi^{t+1}_x = &\sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} \Big) - (1-\omega)\sum_i \left(\sum_{k\neq i} f^t_{k,x-c_i} \right), \\
\phi^{t+1}_x = &\sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} \Big) - (1-\omega)\sum_i \left(\sum_{k\neq i} \left(f^{t-1}_{k,x-c_i-c_k} - \omega \left(f^{t-1}_{k,x-c_i-c_k} - f^{eq,t-1}_{k,x-c_i-c_k} \right) \right) \right),
\end{aligned}$$

doing the same initial procedure to the terms $f^{t-1}_{k,x-c_i}$:

$$\begin{aligned}
\phi^{t+1}_x = &\sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} \Big) - (1-\omega)\sum_i \left(\sum_{k\neq i} \left(f^{t-1}_{k,x-c_i-c_k} - \omega \left(f^{t-1}_{k,x-c_i-c_k} - f^{eq,t-1}_{k,x-c_i-c_k} \right) \right) \right), \\
\phi^{t+1}_x = &\sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} \Big) - (1-\omega)\sum_i \left(\sum_{k\neq i} \left(\omega w_k \phi^{t-1}_{x-c_i-c_k} + (1-\omega) f^{t-1}_{k,x-c_i-c_j} \right) \right),
\end{aligned}$$

notice that $\sum_i \left(\sum_{k\neq i} \left(\omega w_k \phi^{t-1}_{x-c_i-c_k} \right) \right) = \sum_i \left(\sum_{k\neq i} \left(w_k \right) \omega \phi^{t-1}_{x+c_i} \right)$ and using Eq. (6.3) we have $\sum_i \left(\sum_{k\neq i} \left(f^{t-1}_{k,x-c_i-c_j} \right) \right) = \sum_i \left(\phi^{t-1}_{x+c_i} - f^{t-1}_{i,x+c_i} \right)$. So rewriting the above equation base in the observations:

$$\phi^{t+1}_x = \sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} \Big) - (1-\omega)\sum_i \left(\left(1-\omega+\omega \sum_{k\neq i} (w_k) \right) \phi^{t-1}_{x+c_i} \right) + (1-\omega)^2 \sum_i \left(f^{t-1}_{i,x+c_i} \right). \tag{6.5}$$

Interpreting $f^{t-1}_{i,x+c_i}$ with Eq. (6.4):

$$f^{t-1}_{i,x+c_i} = f^{t-2}_{i,x} - \omega \left(f^{t-2}_{i,x} - f^{eq,t-2}_{i,x} \right),$$

and replaing it in Eq. (6.5), we obtained a four-level explicity scheme:

$$\begin{aligned}
\phi^{t+1}_x = &\sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} \Big) - (1-\omega)\sum_i \left(\left(1-\omega+\omega \sum_{k\neq i} (w_k) \right) \phi^{t-1}_{x+c_i} \right) \\
&+(1-\omega)^2 \sum_i \left(f^{t-2}_{i,x} - \omega \left(f^{t-2}_{i,x} - f^{eq,t-2}_{i,x} \right) \right), \\
\phi^{t+1}_x = &\sum_i \Big((1-\omega+\omega w_i)\phi^t_{x-c_i} \Big) - (1-\omega)\sum_i \left(\left(1-\omega+\omega \sum_{k\neq i} (w_k) \right) \phi^{t-1}_{x+c_i} \right) + (1-\omega)^2 \phi^{t-2}_x
\end{aligned} \tag{6.6}$$

Expanding the sums:

$$\begin{aligned}
\phi^{t+1}_x = (1-\omega+\omega w_0)\phi^t_x + (1-\omega+\omega w_1)\phi^t_{x-1} + (1-\omega+\omega w_2)\phi^t_{x+1} - (1-\omega)^2 \Big(\phi^{t-1}_x + \phi^{t-1}_{x-1} + \phi^{t-1}_{x+1} \Big) \\
-(1-\omega)\omega \Big((w_1+w_2)\phi^{t-1}_x + (w_0+w_1)\phi^{t-1}_{x-1} + (w_0+w_2)\phi^{t-1}_{x+1} \Big) + (1-\omega)^2 \phi^{t-2}_x
\end{aligned}$$

6.2 1st Benchmark: Decay of Sinoidal Wave

Considering a 1D **convection–diffusion equation** with constant coefficients

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \qquad x \in \mathbb{R}, t \geq 0,$$

with **sinusoidal initial condition**

$$u(x, 0) = A\sin(kx + \phi),$$

where A is the initial amplitude, k the wavenumber, ϕ a phase shift.

Then the analytical solution is

$$\boxed{u(x,t) = Ae^{-\nu k^2 t}\sin\left(k(x - ct) + \phi\right)}$$

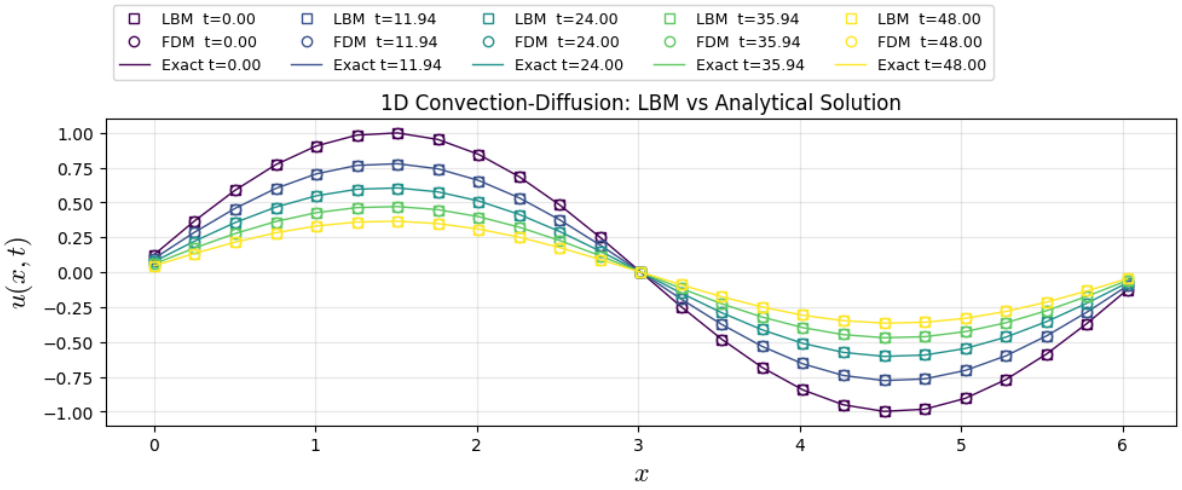
So:

- **Convection** shifts the wave: $x \mapsto x - ct$ (wave travels at speed c).
- **Diffusion** damps the amplitude exponentially: $A(t) = Ae^{-\nu k^2 t}$.

If you use a cosine initial condition $u(x, 0) = A\cos(kx + \phi)$, the solution is the same form:

$$u(x,t) = Ae^{-\nu k^2 t}\cos\left(k(x - ct) + \phi\right).$$

6.2.1 LBM D1Q3 Code - FDM Code (Only Diffusive Case $c = 0$)



6.3 2nd Benchmark: Temperature ramp

Analyzing the accuracy of diffusive equation solver, we apply it to a one-dimensional analytical case. The geometry is described by a domain of length L initialized with a constant temperatura $T(x, t = 0) = 0$. The boundary conditions are given $T(x = 0, t) = 1$ and $T(x = L, t) = 0$.

6.3.1 Steady-State Analytical Solution

The analytical solution of this problem is given by:

$$T(x, t \to \infty) = -\frac{x}{L} + 1.$$

6.3.2 Transient Analytical Solution

For the 1D heat equation $\partial_t T = \nu\, \partial_{xx} T$ on $0 < x < L$ with

$$T(0,t) = 1, \qquad T(L,t) = 0,$$

the steady state is $T_{\text{ss}}(x) = 1 - \frac{x}{L}$. With $u = T - T_{\text{ss}}$ (so $u(0,t) = u(L,t) = 0$), the solution is

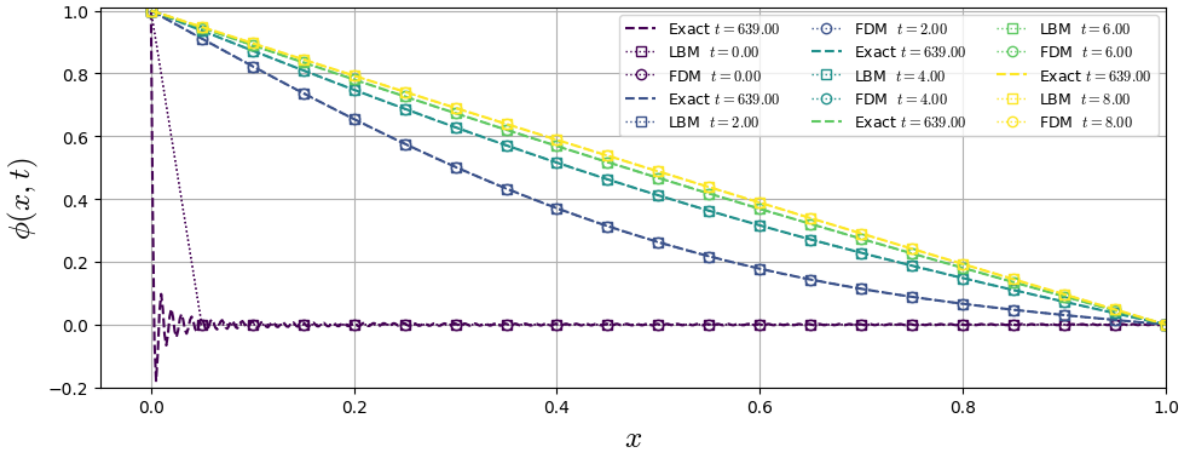
$$T(x,t) = 1 - \frac{x}{L} + \sum_{n=1}^{\infty} B_n\, e^{-\nu(n\pi/L)^2 t}\, \sin\!\Big(\frac{n\pi x}{L}\Big), \quad B_n = \frac{2}{L}\int_0^L \Big[T_0(\xi) - \Big(1 - \frac{\xi}{L}\Big)\Big]\, \sin\!\Big(\frac{n\pi\xi}{L}\Big)\, d\xi.$$

Special case $T_0(x) = 0$: $B_n = -\dfrac{2}{n\pi}$, so

$$T(x,t) = 1 - \frac{x}{L} - \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{1}{n}\, e^{-\nu(n\pi/L)^2 t}\, \sin\!\Big(\frac{n\pi x}{L}\Big).$$

Here's a ready-to-run Python snippet (with a single plot of $T(x,t)$ vs x for multiple times):

6.3.3 LBM D1Q3 Code - FDM Code



6.4 Convergence Errors: Fourth-order Diffusive Equation

The finite-difference interpretation of the lattice Boltzmann equation allows for a precise analysis of the numerical errors associated with the spatial and temporal discretization of the method. Following the procedure presented below, these numerical errors can be identified at different orders of Δx and Δt and, when possible, eliminated through an appropriate choice of model parameters.

Expanding the sums:

$$\phi_x^{t+1} = (1-\omega+\omega w_0)\phi_x^t + (1-\omega+\omega w_1)\phi_{x-1}^t + (1-\omega+\omega w_2)\phi_{x+1}^t - (1-\omega)^2\Big(\phi_x^{t-1} + \phi_{x-1}^{t-1} + \phi_{x+1}^{t-1}\Big)$$
$$-(1-\omega)\omega\Big((w_1+w_2)\phi_x^{t-1} + (w_0+w_1)\phi_{x-1}^{t-1} + (w_0+w_2)\phi_{x+1}^{t-1}\Big) + (1-\omega)^2\phi_x^{t-2}$$

Considering $\alpha_1 = \Omega + a\omega, \quad \alpha_2 = \Omega + (1-2a)\omega, \quad \beta_1 = -\Omega(\Omega + (1-a)\omega), \quad \text{and} \quad \beta_2 = -\Omega(\Omega + 2a\omega),$

where $\Omega = (1-\omega)$ and $a = w_1 = w_2$. Replacing in the Eq. (6.6):

$$\phi_x^{t+1} = \alpha_1\phi_{x-1}^t + \alpha_2\phi_x^t + \alpha_1\phi_{x+1}^t + \beta_1\phi_{x-1}^{t-1} + \beta_2\phi_x^{t-1} + \beta_1\phi_{x+1}^{t-1} + (1-\omega)^2\phi_x^{t-2}$$

To recover the partial differetial equation described by the above equation, we expand each ϕ term in Taylor series:

$$\begin{aligned}
\phi_x^{t\pm1} = &\quad \phi \pm \Delta_t\frac{\partial\phi}{\partial t} + \frac{\Delta_t^2}{2}\frac{\partial^2\phi}{\partial t^2} \pm \frac{\Delta_t^3}{6}\frac{\partial^3\phi}{\partial t^3} + \frac{\Delta_t^4}{24}\frac{\partial^4\phi}{\partial t^4} \pm \mathcal{O}(\Delta t)^4,\\
\phi_{x\pm1}^t = &\quad \phi \pm \Delta_x\frac{\partial\phi}{\partial x} + \frac{\Delta_x^2}{2}\frac{\partial^2\phi}{\partial x^2} \pm \frac{\Delta_x^3}{6}\frac{\partial^3\phi}{\partial x^3} + \frac{\Delta_x^4}{24}\frac{\partial^4\phi}{\partial x^4} \pm \mathcal{O}(\Delta x)^4,\\
\phi_x^{t-2} = &\quad \phi - 2\Delta_t\frac{\partial\phi}{\partial t} + \frac{4\Delta_t^2}{2}\frac{\partial^2\phi}{\partial t^2} - \frac{8\Delta_t^3}{6}\frac{\partial^3\phi}{\partial t^3} + \frac{16\Delta_t^4}{24}\frac{\partial^4\phi}{\partial t^4} \pm \mathcal{O}(\Delta t)^4,\\
\phi_{x\pm1}^{t-1} = &\quad \phi \pm \Delta_x\frac{\partial\phi}{\partial x} - \Delta_t\frac{\partial\phi}{\partial t} + \frac{\Delta_x^2}{2}\frac{\partial^2\phi}{\partial x^2} \mp \Delta_t\Delta_x\frac{\partial^2\phi}{\partial x\partial t} + \frac{\Delta_t^2}{2}\frac{\partial^2\phi}{\partial t^2} \pm \frac{\Delta_x^3}{6}\frac{\partial^3\phi}{\partial x^3}\\
&\mp\frac{\Delta_x^2\Delta_t}{2}\frac{\partial^3\phi}{\partial x^2\partial t} \pm \frac{\Delta_x\Delta_t^2}{2}\frac{\partial^3\phi}{\partial x\partial t^2} - \frac{\Delta_t^3}{6}\frac{\partial^3\phi}{\partial t^3} + \mathcal{O}(\Delta x\Delta t)^3.
\end{aligned} \tag{6.7}$$

Replacing the expanded terms and simplifying the term (this link provide a sympy code that reach the above equation appendix/sympy-code-fdm-diffusive-expansion), we have:

$$0 = \Delta_t\omega^2\frac{\partial\phi}{\partial t} + \Delta_x^2 a\omega\left(\omega-2\right)\frac{\partial^2\phi}{\partial x^2} + \frac{\Delta_t^2\omega\left(4-3\omega\right)}{2}\frac{\partial^2\phi}{\partial t^2} - \Delta_t\Delta_x^2\left(a\omega^2 - a\omega - \omega + 1\right)\frac{\partial^3\phi}{\partial x^2\partial t} + \frac{\Delta_x^4 a\omega\left(\omega-2\right)}{12}\frac{\partial^4\phi}{\partial x^4}$$

$$\Delta_t\omega^2\frac{\partial\phi}{\partial t} = -\Delta_x^2 a\omega\left(\omega-2\right)\frac{\partial^2\phi}{\partial x^2} - \frac{\Delta_t^2\omega\left(4-3\omega\right)}{2}\frac{\partial^2\phi}{\partial t^2} + \Delta_t\Delta_x^2\left(a\omega^2 - a\omega - \omega + 1\right)\frac{\partial^3\phi}{\partial x^2\partial t} - \frac{\Delta_x^4 a\omega\left(\omega-2\right)}{12}\frac{\partial^4\phi}{\partial x^4}$$

$$\frac{\partial\phi}{\partial t} = \frac{a\left(2-\omega\right)}{\omega}\frac{\Delta_x^2}{\Delta_t}\frac{\partial^2\phi}{\partial x^2} + \frac{\left(3\omega-4\right)\Delta_t}{2\omega}\frac{\partial^2\phi}{\partial t^2} + \frac{\left(a\omega^2 - a\omega - \omega + 1\right)\Delta_x^2}{\omega^2}\frac{\partial^3\phi}{\partial x^2\partial t} + \frac{a\left(2-\omega\right)}{12\omega}\frac{\Delta_x^4}{\Delta_t}\frac{\partial^4\phi}{\partial x^4}$$

where $\nu = \frac{a(2-\omega)}{\omega}\frac{\Delta_x^2}{\Delta_t}$ is the same term recovered in the Chapman-Enskog Analysis. To enforce fourth-order Δ_x convergence rate, employ the relations

$$\frac{\partial\phi}{\partial t} = \nu\frac{\partial^2\phi}{\partial x^2}, \qquad \frac{\partial^2\phi}{\partial t^2} = \nu^2\frac{\partial^4\phi}{\partial x^4}, \qquad \frac{\partial^3\phi}{\partial x^2\partial t} = \nu\frac{\partial^4\phi}{\partial x^4}$$

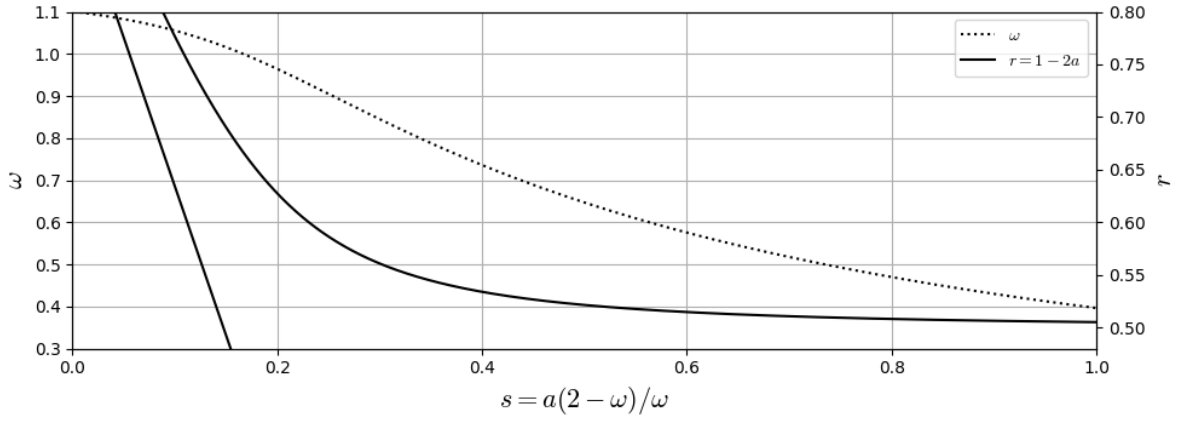
$$\frac{\partial\phi}{\partial t} = \nu\frac{\partial^2\phi}{\partial x^2} + \frac{a\left(2-\omega\right)}{\omega}\frac{\left(3\omega-4\right)\nu\Delta_x^2}{2\omega}\frac{\partial^4\phi}{\partial x^4} + \frac{\left(a\omega^2 - a\omega - \omega + 1\right)\nu\Delta_x^2}{\omega^2}\frac{\partial^4\phi}{\partial x^4} + \frac{\nu\Delta_x^2}{12}\frac{\partial^4\phi}{\partial x^4}$$

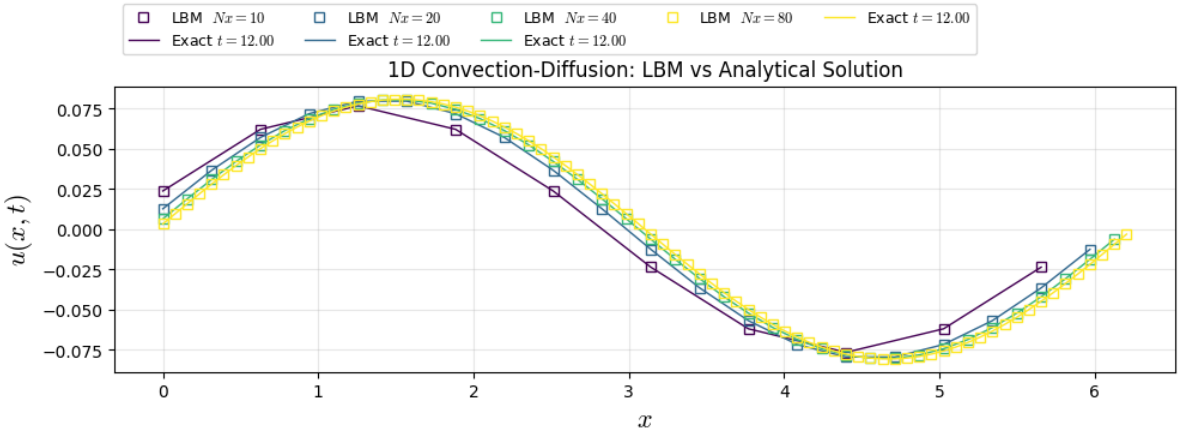
$$\frac{\partial\phi}{\partial t} = \nu\frac{\partial^2\phi}{\partial x^2} + \frac{a\left(6\omega-8-3\omega^2+4\omega\right)\nu\Delta_x^2}{2\omega^2}\frac{\partial^4\phi}{\partial x^4} + \frac{\left(a\omega^2 - a\omega - \omega + 1\right)\nu\Delta_x^2}{\omega^2}\frac{\partial^4\phi}{\partial x^4} + \frac{\omega^2\nu\Delta_x^2}{12\omega^2}\frac{\partial^4\phi}{\partial x^4}$$

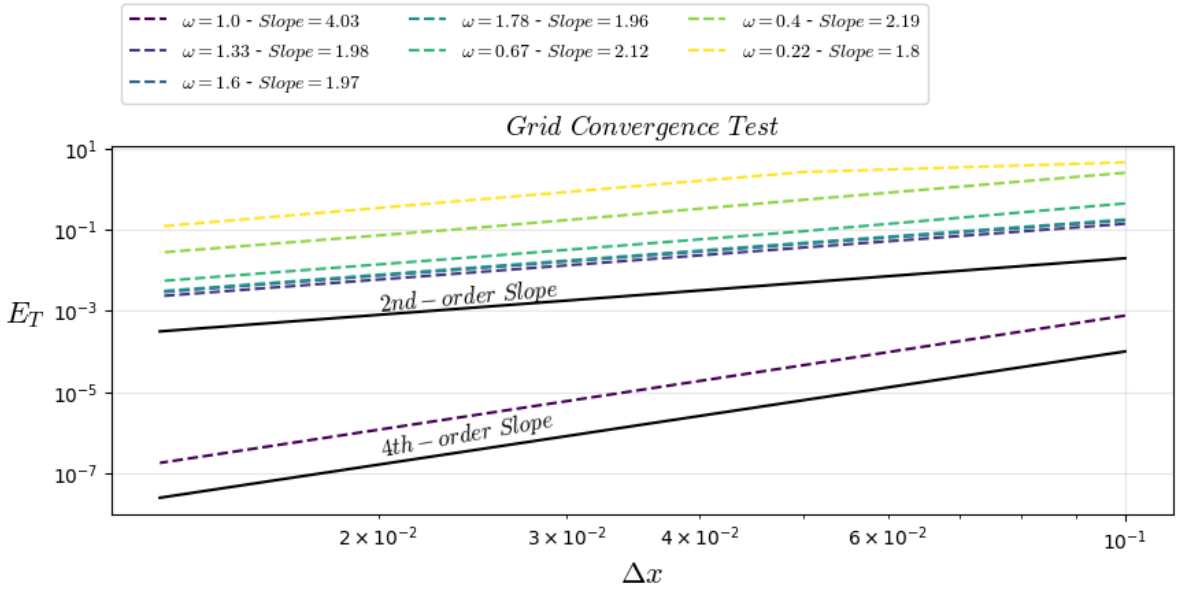
$$\frac{\partial\phi}{\partial t} = \nu\frac{\partial^2\phi}{\partial x^2} + \frac{\nu\Delta_x^2}{2\omega^2}\left(a\left(10\omega-8-3\omega^2\right) + 2\left(a\omega^2 - a\omega - \omega + 1\right) + \frac{\omega^2}{6}\right)\frac{\partial^4\phi}{\partial x^4}$$

$$\frac{\partial\phi}{\partial t} = \nu\frac{\partial^2\phi}{\partial x^2} + \frac{\nu\Delta_x^2}{2\omega^2}\left(a\left(8\omega-8-\omega^2\right) + 2\left(1-\omega\right) + \frac{\omega^2}{6}\right)\frac{\partial^4\phi}{\partial x^4}$$

$$\frac{\partial\phi}{\partial t} = \nu\frac{\partial^2\phi}{\partial x^2} - \frac{\nu\Delta_x^2}{2\omega^2}\left(a\left(\omega^2-8\omega+8\right) - \frac{\omega^2-12\omega+12}{6}\right)\frac{\partial^4\phi}{\partial x^4} + \mathcal{O}(\Delta x)^4$$


ω
$r = 1 - 2a$
r
$s = a(2 - \omega)/\omega$


LBM $Nx = 10$
LBM $Nx = 20$
LBM $Nx = 40$
LBM $Nx = 80$
Exact $t = 12.00$
1D Convection-Diffusion: LBM vs Analytical Solution
$u(x,t)$
x


$\omega = 1.0$ - $Slope = 4.03$
$\omega = 1.33$ - $Slope = 1.98$
$\omega = 1.6$ - $Slope = 1.97$
$\omega = 1.78$ - $Slope = 1.96$
$\omega = 0.67$ - $Slope = 2.12$
$\omega = 0.4$ - $Slope = 2.19$
$\omega = 0.22$ - $Slope = 1.8$
Grid Convergence Test
E_T
$2nd - order\ Slope$
$4th - order\ Slope$
Δx

6.5 Extension to Lattices with d-Dimensions and q-Velocities ($DdQq$):

Through a general mathematical manipulation of the single relaxation time lattice Boltzmann equation, it is possible to derive a general expansion that expresses the lattice framework in a finite-difference form.

Building on the earlier analysis presented in Suga [Sug10], Suga [Sug14] proposed a generalized formulation for diffusive transport on $DdQq$ lattices, considering both isotropic and anisotropic diffusion. The resulting equations are presented in Eq. (6.8), with truncation-error terms highlighted in red and retained through fourth order in Δx. Notice that additional second-order errors may arise when extending the formulation to higher dimensions:

$$
\begin{aligned}
D1Q3-Isotropic: \quad \frac{\partial \phi}{\partial t} &= \nu \frac{\partial^2 \phi}{\partial x^2} - \frac{\nu \Delta_x^2}{2\omega^2}\left(a\left(\omega^2-8\omega+8\right)-\frac{\omega^2-12\omega+12}{6}\right)\frac{\partial^4 \phi}{\partial x^4}+\mathcal{O}(\Delta x)^4 \\
D2Q9-Isotropic: \quad \frac{\partial \phi}{\partial t} &= \nu\left(\frac{\partial^2 \phi}{\partial x^2}+\frac{\partial^2 \phi}{\partial y^2}\right)-\Delta_x^2 R1\left(\frac{\partial^4 \phi}{\partial x^4}+\frac{\partial^4 \phi}{\partial y^4}\right)-\Delta_x^2 R12\left(\frac{\partial^4 \phi}{\partial x^2 \partial y^2}\right)+\mathcal{O}(\Delta x)^4 \\
D2Q13-Anisotropic: \quad \frac{\partial \phi}{\partial t} &= \nu_x \frac{\partial^2 \phi}{\partial x^2}+\nu_y \frac{\partial^2 \phi}{\partial y^2}-\Delta_x^2 R1\frac{\partial^4 \phi}{\partial x^4}+\Delta_x^2 R2\frac{\partial^4 \phi}{\partial y^4}-\Delta_x^2 R12\left(\frac{\partial^4 \phi}{\partial x^2 \partial y^2}\right)+\mathcal{O}(\Delta x)^4 \\
D3Q19-Isotropic: \quad \frac{\partial \phi}{\partial t} &= \nu\left(\frac{\partial^2 \phi}{\partial x^2}+\frac{\partial^2 \phi}{\partial y^2}+\frac{\partial^2 \phi}{\partial z^2}\right)-\Delta_x^2 R1\left(\frac{\partial^4 \phi}{\partial x^4}+\frac{\partial^4 \phi}{\partial y^4}+\frac{\partial^4 \phi}{\partial z^4}\right) \\
&\quad -\Delta_x^2 R12\left(\frac{\partial^4 \phi}{\partial x^2 \partial y^2}+\frac{\partial^4 \phi}{\partial x^2 \partial z^2}+\frac{\partial^4 \phi}{\partial y^2 \partial z^2}\right)+\mathcal{O}(\Delta x)^4 \\
D3Q25-Anisotropic: \quad \frac{\partial \phi}{\partial t} &= \nu_x \frac{\partial^2 \phi}{\partial x^2}+\nu_y \frac{\partial^2 \phi}{\partial y^2}+\nu_z \frac{\partial^2 \phi}{\partial z^2}-\Delta_x^2\left(R1\frac{\partial^4 \phi}{\partial x^4}+R2\frac{\partial^4 \phi}{\partial z^4}+R3\frac{\partial^4 \phi}{\partial y^4}\right) \\
&\quad -\Delta_x^2\left(R12\frac{\partial^4 \phi}{\partial x^2 \partial y^2}+R13\frac{\partial^4 \phi}{\partial x^2 \partial z^2}+R23\frac{\partial^4 \phi}{\partial y^2 \partial z^2}\right)+\mathcal{O}(\Delta x)^4
\end{aligned}
\tag{6.8}
$$

In the Section XX we will present a generalized expansion of LB equation with single relaxation time in FDM framewrok for any equilibrium distribution function with n conservative terms in $DdQq$ lattice structures.

6.6 Diffusive Equation Interpretation with Multi-Relaxation Time (MRT-Framework)

The multiple-relaxation-time (MRT) collision model is a well-established scheme in the LBM literature that allows different non-equilibrium moments to relax independently. This flexibility enables various improvements in the numerical representation of partial differential equations. In the present section, we extend the FDM interpretation to the MRT framework of diffusive equation, following the analysis presented by Lin *et al.* [LHSC21].

Given the LB equation with MRT for the diffusive equation:

$$
f_i(x_\alpha + e_{i,\alpha}\delta t, t+\delta t) = f_i(x_\alpha, t) - (\mathbf{M}^{-1}\mathbf{S}\mathbf{M})_{ik}\left(f_k - f_k^{eq}\right), \tag{6.9}
$$

where $\mathbf{M}$ is matrix of moments, $\mathbf{S}$ is the diagonal matrix of relaxation times, and $\mathbf{M}^{-1}$ is the inverse matrix of $\mathbf{M}$. Considering the problem description in a lattice D1Q3, we have following a Gram-Schmidt procedure:

$$
\mathbf{M} = \begin{pmatrix} 1 \\ e_{i,x} \\ 3e_{i,x}^2 - 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ -2 & 1 & 1 \end{pmatrix}, \qquad \mathbf{S} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & s_1 & 0 \\ 0 & 0 & s_2 \end{pmatrix}, \qquad \mathbf{M}^{-1} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ \frac{1}{3} & -\frac{1}{2} & \frac{1}{6} \end{pmatrix}.
$$

Applying the same shift scheme employed in Eq. (6.4) to Eq. (6.9), we obtain

$$\begin{bmatrix} f_{0,x}^{t+1} \\ f_{1,x}^{t+1} \\ f_{2,x}^{t+1} \end{bmatrix} = \begin{bmatrix} \phi_x^t s_2 w_0 - \frac{\phi_x^t s_2}{3} - \frac{2f_{0,x}^t s_2}{3} + f_{0,x}^t + \frac{f_{1,x}^t s_2}{3} + \frac{f_{2,x}^t s_2}{3} \\ -\frac{\phi_{x-1}^t s_2 w_0}{2} + \frac{\phi_{x-1}^t s_2}{6} + \frac{f_{0,x-1}^t s_2}{3} - \frac{f_{1,x-1}^t s_1}{2} - \frac{f_{1,x-1}^t s_2}{6} + f_{1,x-1}^t + \frac{f_{2,x-1}^t s_1}{2} - \frac{f_{2,x-1}^t s_2}{6} \\ -\frac{\phi_{x+1}^t s_2 w_0}{2} + \frac{\phi_{x+1}^t s_2}{6} + \frac{f_{0,x+1}^t s_2}{3} + \frac{f_{1,x+1}^t s_1}{2} - \frac{f_{1,x+1}^t s_2}{6} - \frac{f_{2,x+1}^t s_1}{2} - \frac{f_{2,x+1}^t s_2}{6} + f_{2,x+1}^t \end{bmatrix},$$

expanding each term and replacing the conservative terms:

$$f_{0,x}^{t+1} = \phi_x^t s_2 w_0 + f_{0,x}^t \left(1 - s_2\right),$$

$$f_{1,x}^{t+1} = -\frac{\phi_{x-1}^t s_2 w_0}{2} + \frac{f_{0,x-1}^t s_2}{2} + f_{1,x-1}^t + \frac{s_1\left(-f_{1,x-1}^t + f_{2,x-1}^t\right)}{2},$$

$$f_{2,x}^{t+1} = -\frac{\phi_{x+1}^t s_2 w_0}{2} + \frac{f_{0,x+1}^t s_2}{2} + f_{2,x+1}^t + \frac{s_1\left(f_{1,x+1}^t - f_{2,x+1}^t\right)}{2}.$$

Summing over all lattice direction, we have:

$$\begin{aligned} \phi_{x+1}^t = f_{1,x-1}^t + f_{2,x+1}^t + f_{0,x}^t \left(1 - s_2\right) + s_2 \left(\frac{f_{0,x+1}^t}{2} + \frac{f_{0,x-1}^t}{2}\right) \\ + \frac{s_1\left(f_{1,x+1}^t - f_{1,x-1}^t - f_{2,x+1}^t + f_{2,x-1}^t\right)}{2} + s_2 w_0 \left(\phi_x^t - \frac{\phi_{x+1}^t}{2} - \frac{\phi_{x-1}^t}{2}\right) \end{aligned},$$

Using the shift interpretation for the distribution functions, we obtain

$$f_{0,x}^t + f_{1,x+1}^t + f_{2,x-1}^t = \phi_x^{t-1}, \tag{6.10}$$

applying this relation to the equation above:

$$\begin{aligned} \phi_{x+1}^t = f_{1,x-1}^t + f_{2,x+1}^t + f_{0,x}^t \left(1 - s_2\right) + s_2 \left(\frac{f_{0,x+1}^t}{2} + \frac{f_{0,x-1}^t}{2}\right) \\ + \frac{s_1\left(\phi_x^{t-1} - f_{0,x}^t - f_{1,x-1}^t - f_{2,x+1}^t\right)}{2} + s_2 w_0 \left(\phi_x^t - \frac{\phi_{x+1}^t}{2} - \frac{\phi_{x-1}^t}{2}\right) \end{aligned}.$$

To repalce the term $f_{0,x}^t + f_{1,x-1}^t + f_{2,x+1}^t$, in the above equation we use the $\sum_i f_{i,x}^t = \phi_x^t$:

$$f_{0,x}^t + f_{1,x-1}^t + f_{2,x+1}^t = \phi_{x+1}^t + \phi_{x-1}^t - \phi_x^{t-1} - f_{0,x+1}^t - f_{0,x-1}^t + 2f_{0,x}^t,$$

where

$$\begin{aligned} \phi_{x+1}^t \quad &= \phi_x^t s_2 w_0 + \phi_{x+1}^t \left(-\frac{s_1}{2} - \frac{s_2 w_0}{2} + 1\right) + \phi_{x-1}^t \left(-\frac{s_1}{2} - \frac{s_2 w_0}{2} + 1\right) + \phi_x^{t-1}\left(s_1 - 1\right) \\ &+ \left(f_{0,x+1}^t + f_{0,x-1}^t - 2f_{0,x}^t\right)\left(\frac{s_1}{2} + \frac{s_2}{2} - 1\right) \end{aligned} \tag{6.11}$$

To replace the distribution function in the above relation, we manipulate:

$$f^t_{0,x+1} + f^t_{0,x-1} - 2f^t_{0,x} = -2\phi^t_x + f^t_{0,x+1} + f^t_{0,x-1} + 2f^t_{1,x} + 2f^t_{2,x},$$

$$f^t_{0,x+1} + f^t_{0,x-1} - 2f^t_{0,x} = -2\phi^t_x + f^{t-1}_{0,x+1} + f^{t-1}_{0,x-1} + 2f^{t-1}_{1,x-1} + 2f^{t-1}_{2,x+1} + s_1\left(f^{t-1}_{1,x+1} - f^{t-1}_{1,x-1} - f^{t-1}_{2,x+1} + f^{t-1}_{2,x-1}\right),$$

$$\begin{aligned} f^t_{0,x+1} + f^t_{0,x-1} - 2f^t_{0,x} = &-f^{t-1}_{0,x+1} - f^{t-1}_{0,x-1} - 2\phi^t_x - 2f^{t-1}_{1,x+1} - 2f^{t-1}_{2,x-1} + 2\phi^{t-1}_{x+1} + 2\phi^{t-1}_{x-1} \\ &+ s_1\left(-\phi^{t-1}_{x+1} - \phi^{t-1}_{x-1} + f^{t-1}_{0,x+1} + f^{t-1}_{0,x-1} + 2f^{t-1}_{1,x+1} + 2f^{t-1}_{2,x-1}\right) \end{aligned},$$

$$f^t_{0,x+1} + f^t_{0,x-1} - 2f^t_{0,x} = -2\phi^t_x - \phi^{t-1}_{x+1}(s_1 - 2) - \phi^{t-1}_{x-1}(s_1 - 2) + (s_1 - 1)\left(2\phi^{t-2}_x + f^{t-1}_{0,x+1} + f^{t-1}_{0,x-1} - 2f^{t-1}_{0,x}\right).$$

Replacing the above relation in ϕ^{t+1}_x term:

$$\begin{aligned} \phi^{t+1}_x = &\phi^{t-1}_x(s_1 - 1) + \left(-\frac{\phi^t_{x+1}}{2} - \frac{\phi^t_{x-1}}{2}\right)(s_1 + s_2 w_0 - 2) + \left(-\frac{\phi^{t-1}_{x+1}}{2} - \frac{\phi^{t-1}_{x-1}}{2}\right)(s_1^2 + s_1 s_2 - 4s_1 - 2s_2 + 4) \\ &- \phi^t_x(s_1 - s_2 w_0 + s_2 - 2) + (s_1 - 1)(s_1 + s_2 - 2)\left(\phi^{t-2}_x + \frac{f^{t-1}_{0,x+1}}{2} + \frac{f^{t-1}_{0,x-1}}{2} - f^{t-1}_{0,x}\right) \end{aligned}.$$

Applying a time shift in the Eq. (6.11) to replace $f^{t-1}_{0,x+1}/2 + f^{t-1}_{0,x-1}/2 - f^{t-1}_{0,x}$ in the above equation, we obtain in the final form ϕ^{t+1}_x only dependent of previous moments:

$$\begin{aligned} \phi^{t+1}_x = &\phi^t_x(s_2 w_0 - s_2 + 1) + \phi^t_{x+1}\left(-\frac{s_1}{2} - \frac{s_2 w_0}{2} + 1\right) \\ &+ \phi^{t-1}_{x+1}\left(\frac{s_1 s_2 w_0}{2} - \frac{s_1 s_2}{2} + \frac{s_1}{2} - \frac{s_2 w_0}{2} + s_2 - 1\right) + \phi^t_{x-1}\left(-\frac{s_1}{2} - \frac{s_2 w_0}{2} + 1\right) \\ &+ \phi^{t-1}_{x-1}\left(\frac{s_1 s_2 w_0}{2} - \frac{s_1 s_2}{2} + \frac{s_1}{2} - \frac{s_2 w_0}{2} + s_2 - 1\right) + \phi^{t-1}_x(-s_1 s_2 w_0 + s_1 + s_2 w_0 - 1) \\ &+ \phi^{t-2}_x(s_1 s_2 - s_1 - s_2 + 1) \end{aligned}.$$

6.6.1 Fourth-order MRT Diffusive Equation

Following the same error analysis performed in the previous *section* for the single-relaxation-time model, we expand the relevant terms in Taylor series (Eq. (6.7)) and substitute them into the derived lattice Boltzmann equation to recover the corresponding partial differential equation, including truncation errors up to fourth order. The symbolic derivation leading to this equation is provided in the accompanying SymPy code. Rearranging all terms on one side of the equation, we obtain:

$$\begin{aligned} 0 = &\frac{\Delta_t^2(-3s_1 s_2 + 2s_1 + 2s_2)}{2}\frac{\partial^2\phi}{\partial t^2} + \frac{\Delta_t \Delta_x^2(s_1 s_2 w_0 - s_1 s_2 + s_1 - s_2 w_0 + 2s_2 - 2)}{2}\frac{\partial^3\phi}{\partial x^2 \partial t} \\ &+\Delta_t s_1 s_2\frac{\partial\phi}{\partial t} + \frac{\Delta_x^4 s_2(-s_1 w_0 + s_1 + 2w_0 - 2)}{24}\frac{\partial^4\phi}{\partial x^4} + \frac{\Delta_x^2 s_2(-s_1 w_0 + s_1 + 2w_0 - 2)}{2}\frac{\partial^2\phi}{\partial x^2}. \end{aligned}$$

Isolation the first order partial derivative term in left-had side of equation:

$$\begin{aligned} \Delta_t s_1 s_2\frac{\partial\phi}{\partial t} = &-\frac{\Delta_x^2 s_2(-s_1 w_0 + s_1 + 2w_0 - 2)}{2}\frac{\partial^2\phi}{\partial x^2} - \frac{\Delta_t^2(-3s_1 s_2 + 2s_1 + 2s_2)}{2}\frac{\partial^2\phi}{\partial t^2} \\ &-\frac{\Delta_t \Delta_x^2(s_1 s_2 w_0 - s_1 s_2 + s_1 - s_2 w_0 + 2s_2 - 2)}{2}\frac{\partial^3\phi}{\partial x^2 \partial t} - \frac{\Delta_x^4 s_2(-s_1 w_0 + s_1 + 2w_0 - 2)}{24}\frac{\partial^4\phi}{\partial x^4}, \\ \frac{\partial\phi}{\partial t} = &\frac{\Delta_x^2(1 - w_0)(2 - s_1)}{2\Delta_t s_1}\frac{\partial^2\phi}{\partial x^2} - \frac{\Delta_t(-3s_1 s_2 + 2s_1 + 2s_2)}{2s_1 s_2}\frac{\partial^2\phi}{\partial t^2} \\ &-\frac{\Delta_x^2(s_1 s_2 w_0 - s_1 s_2 + s_1 - s_2 w_0 + 2s_2 - 2)}{2s_1 s_2}\frac{\partial^3\phi}{\partial x^2 \partial t} + \frac{\Delta_x^4(1 - w_0)(2 - s_1)}{24\Delta_t s_1}\frac{\partial^4\phi}{\partial x^4}. \end{aligned}$$

Considering diffusive coeficient $\nu = (1 - w_0)\left(\frac{1}{s_1} - \frac{1}{2}\right)\frac{\Delta_x^2}{\Delta_t}$, we can rewrite the equation

$$\frac{\partial \phi}{\partial t} = \nu \frac{\partial^2 \phi}{\partial x^2} - \frac{\Delta_t \left(-3 s_1 s_2 + 2 s_1 + 2 s_2\right)}{2 s_1 s_2} \frac{\partial^2 \phi}{\partial t^2} - \frac{\Delta_x^2 \left(s_1 s_2 w_0 - s_1 s_2 + s_1 - s_2 w_0 + 2 s_2 - 2\right)}{2 s_1 s_2} \frac{\partial^3 \phi}{\partial x^2 \partial t} + \frac{\nu \Delta_x^2}{12} \frac{\partial^4 \phi}{\partial x^4}$$

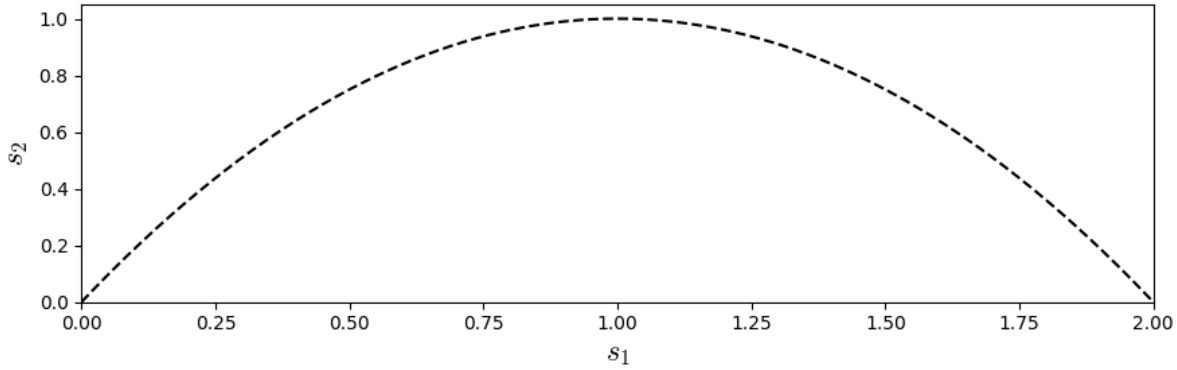
Using the relations:

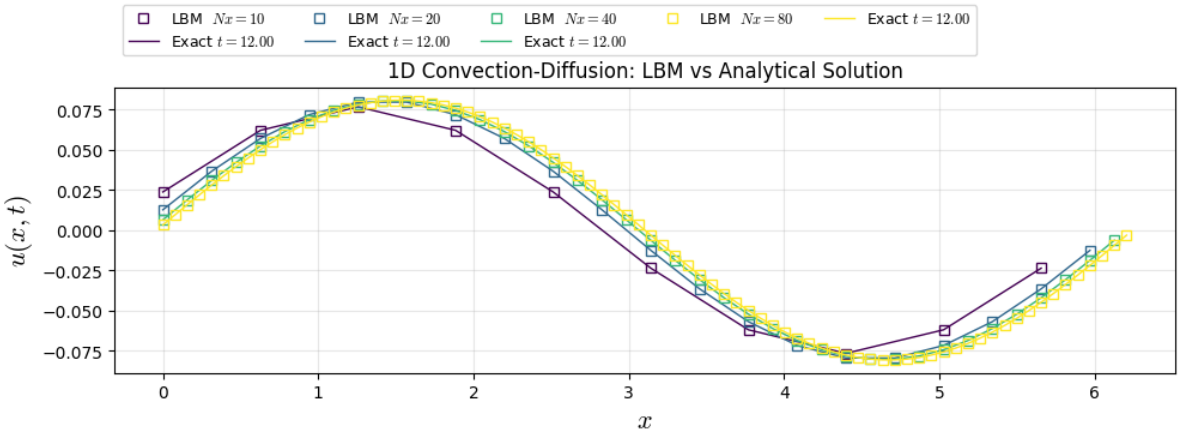
$$\frac{\partial \phi}{\partial t} = \nu \frac{\partial^2 \phi}{\partial x^2}, \qquad \frac{\partial^2 \phi}{\partial t^2} = \nu^2 \frac{\partial^4 \phi}{\partial x^4}, \qquad \frac{\partial^3 \phi}{\partial x^2 \partial t} = \nu \frac{\partial^4 \phi}{\partial x^4},$$

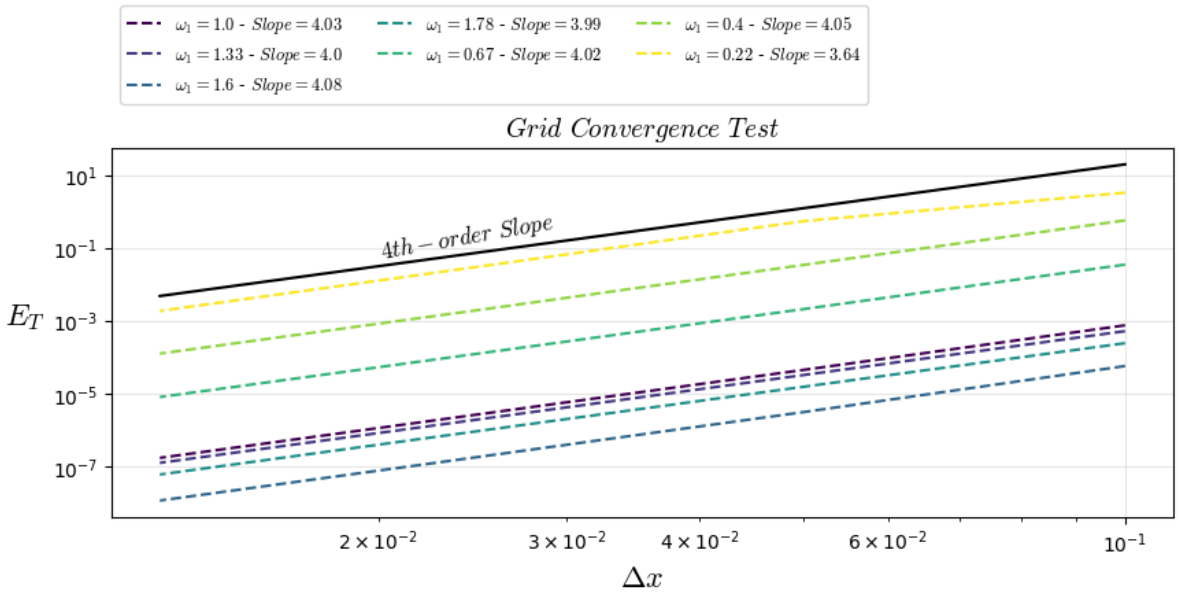
we can reduce the equation in terms of $\frac{\partial^4 \phi}{\partial x^4}$:

$$\begin{aligned}
\frac{\partial \phi}{\partial t} &= \nu \frac{\partial^2 \phi}{\partial x^2} - \frac{\Delta_x^2 (1 - w_0)\left(2 - s_1\right)}{2 s_1} \frac{\nu \left(-3 s_1 s_2 + 2 s_1 + 2 s_2\right)}{2 s_1 s_2} \frac{\partial^4 \phi}{\partial x^4} \\
&\quad - \frac{\Delta_x^2 \nu \left(s_1 s_2 w_0 - s_1 s_2 + s_1 - s_2 w_0 + 2 s_2 - 2\right)}{2 s_1 s_2} \frac{\partial^4 \phi}{\partial x^4} + \frac{\nu \Delta_x^2}{12} \frac{\partial^4 \phi}{\partial x^4} \\
\frac{\partial \phi}{\partial t} &= \nu \frac{\partial^2 \phi}{\partial x^2} + \frac{\Delta_x^2 \nu \left(3 s_1^2 s_2 w_0 - 2 s_1^2 s_2 - 6 s_1^2 w_0 - 18 s_1 s_2 w_0 + 12 s_1 s_2 + 12 s_1 w_0 + 12 s_2 w_0 - 12 s_2\right)}{12 s_1^2 s_2} \frac{\partial^4 \phi}{\partial x^4}
\end{aligned}$$

Making the term $\left(3 s_1^2 s_2 w_0 - 2 s_1^2 s_2 - 6 s_1^2 w_0 - 18 s_1 s_2 w_0 + 12 s_1 s_2 + 12 s_1 w_0 + 12 s_2 w_0 - 12 s_2\right)$ equal to zero e can reach fourth order approximation. For $w_0 = 2/3$ (weight specific from the lattice D1Q3) the relation reduce to $4 s_1^2 - 8 s_1 + 4 s_2$:






$\omega_1 = 1.0$ - $Slope = 4.03$
$\omega_1 = 1.33$ - $Slope = 4.0$
$\omega_1 = 1.6$ - $Slope = 4.08$
$\omega_1 = 1.78$ - $Slope = 3.99$
$\omega_1 = 0.67$ - $Slope = 4.02$
$\omega_1 = 0.4$ - $Slope = 4.05$
$\omega_1 = 0.22$ - $Slope = 3.64$
Grid Convergence Test
$4th - order\ Slope$
E_T
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
2×10^{-2}
3×10^{-2}
4×10^{-2}
6×10^{-2}
10^{-1}
Δx

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